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Biology
Class 12

Sindh Textbook Board

(English Medium)

Based on the Sindh Textbook Board Book

Chapter 1
Homeostasis

Chapter 2
Support and Movement

Chapter 3
Nervous Coordination

Chapter 4
Chemical Coordination

Chapter 5
Animal Behaviour

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Reproduction

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Development and Aging

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Homeostasis

CHAPTER 1

Homeostasis and Feedback Mechanisms

LESSON 1

1. Homeostasis

Homeostasis is a self-regulating process by which biological systems maintain stability while adjusting to the changing conditions. It is the overall process of keeping the internal environment stable. Osmoregulation and excretion are only parts of it that help in maintaining it.

2. Why physiological integration of the internal environment is important

Cells can work properly only when temperature, pH, water, salts and nutrients around them stay within narrow limits. Physiological integration coordinates organs such as the kidney, liver, lungs and endocrine glands so that this internal environment remains stable. Without it, enzymes would fail and the organism could not survive changes in the external environment.

3. Negative feedback and positive feedback

- **Negative feedback:** counteracts the change and restores the stable condition. It keeps most body variables stable by reversing any change.
- **Positive feedback:** does not maintain a stable, homeostatic condition; rather it intensifies the change that is happening to the body. It amplifies the original change instead of reversing it, and rapidly completes certain processes such as blood clotting and childbirth.

Feedback	Effect on change	Role
Negative feedback	Counteracts and reverses the change	Keeps most body variables stable
Positive feedback	Amplifies and intensifies the change	Rapidly completes processes such as blood clotting and childbirth

4. Balance between the two feedback mechanisms

It is important that positive and negative feedback mechanisms counteract upon each other. Negative feedback keeps most body variables stable by reversing any change, while positive feedback rapidly completes certain processes such as blood clotting and childbirth. The two must balance and counteract each other so that a useful positive response does not continue endlessly and upset homeostasis. This balance allows both stability and timely, controlled change in the body.

Osmoregulation in Animals

LESSON 2

1. Osmoregulation

Osmoregulation is a part of homeostasis that helps in maintaining it by keeping the internal osmotic concentration constant. Aquatic osmoregulators overcome their osmoregulatory problems by excretion and active ion transport.

2. Ionocytes in freshwater organisms

Fresh water organisms have specialized cells located in their gills and in skin which actively extract Na^+, Cl^- and Ca^+ from external medium and excrete H^+ or basic HCO_3^- for acid base balance in the body fluids. These cells are called ionocytes.

- **Ionocytes:** the specialized ion-transporting cells of gills and skin in freshwater animals. They take up needed ions from the dilute surroundings and remove H^+ or HCO_3^- to keep acid-base balance.

3. How aquatic osmoregulators overcome osmoregulatory problems

- **Freshwater animals:** face continuous entry of water and loss of salts, so they excrete large amounts of dilute urine and actively absorb Na^+, Cl^- and other ions through ionocytes in the gills and skin.
- **Marine animals:** face the opposite problem of water loss and salt gain, so they drink seawater, excrete excess salt through gills and produce very little concentrated urine.

Animal group	Problem	Response
Freshwater animals	Continuous entry of water and loss of salts	Excrete large amounts of dilute urine; actively absorb Na+, Cl- and other ions through ionocytes in the gills and skin
Marine animals	Water loss and salt gain	Drink seawater, excrete excess salt through gills and produce very little concentrated urine

In both cases excretion and active ion transport keep the internal osmotic concentration constant.

Excretion and Nitrogenous Wastes

LESSON 3

1. Excretion

Excretion is a part of homeostasis that helps in maintaining it. Animals excrete different types of excretory waste, and the type of nitrogenous waste depends on the availability of water and the habitat of the animal.

2. Why animals excrete different types of excretory waste

- **Ammonia:** highly toxic but needs much water for its removal, so it is excreted mainly by aquatic animals.
- **Urea:** less toxic and needs moderate water, so it is excreted by mammals.
- **Uric acid:** almost non-toxic and needs very little water, so it suits birds, reptiles and desert animals.

Nitrogenous waste	Toxicity	Water needed for removal	Excreted mainly by
Ammonia	Highly toxic	Much water	Aquatic animals
Urea	Less toxic	Moderate water	Mammals
Uric acid	Almost non-toxic	Very little water	Birds, reptiles and desert animals

The type of nitrogenous waste depends on the availability of water and the habitat of the animal.

The Human Kidney and Structure of the Nephron

LESSON 4

1. The nephron

A nephron is the structural and functional unit of the kidney, and each kidney contains about a million nephrons. It begins with the renal corpuscle, made of the glomerulus, a tuft of capillaries, surrounded by the cup-shaped Bowman's capsule whose inner layer is formed of podocytes; here ultrafiltration of blood takes place. The filtrate then passes through the proximal convoluted tubule, where most glucose, amino acids, salts and water are reabsorbed, then through the loop of Henle, whose descending and ascending limbs create the counter current gradient, and then through the distal convoluted tubule, where secretion of H+, K+ and drugs and further reabsorption occur. Finally the nephron opens into the collecting duct, which carries urine to the renal pelvis while reabsorbing water under ADH control.

- **Renal corpuscle:** made of the glomerulus, a tuft of capillaries, surrounded by the cup-shaped Bowman's capsule; ultrafiltration of blood takes place here.
- **Proximal convoluted tubule:** most glucose, amino acids, salts and water are reabsorbed here.
- **Loop of Henle:** its descending and ascending limbs create the counter current gradient.
- **Distal convoluted tubule:** secretion of H+, K+ and drugs and further reabsorption occur here.
- **Collecting duct:** carries urine to the renal pelvis while reabsorbing water under ADH control.

2. Podocytes and filtration

Podocyte cells are specialized cells of the glomerulus that are wrapped around blood capillaries and play an active role in preventing plasma proteins from entering the urinary ultrafiltrate. Podocytes wrap the glomerular capillaries with their foot processes and form part of the filtration barrier. This barrier allows water and small solutes to pass but holds back plasma proteins and blood cells. The inner layer of Bowman's capsule is formed of podocytes.

3. Glomerular filtration rate

Glomerular filtration rate (GFR) is directly proportional to the hydrostatic pressure exerted in glomerulus wall. When hydrostatic pressure in the glomerulus rises, more fluid is forced through the filtration membrane, so GFR increases. When the pressure falls, GFR also falls.

4. Distal convoluted tubule and acid-base balance

The distal convoluted tubule secretes ions such as hydrogen, potassium, and NH3 into the filtrate while reabsorbing the HCO3- from the filtrate. It secretes H+, K+ and ammonia into the filtrate and reabsorbs bicarbonate from it. In this way it plays an important role in acid-base and electrolyte balance.

5. Counter current mechanism in kidney

The counter current mechanism in the loop of Henle builds a strong osmotic gradient in the medulla, from the cortex towards the inner medulla. This gradient allows water to be drawn out of the collecting duct under the control of ADH. As a result, the kidney can produce urine more concentrated than blood and conserve water when the body needs it.

6. Urea and the osmotic gradient in inner medulla

Urea is another compound that increases the osmotic gradient in inner medulla. Along with NaCl, urea accumulates in the inner medulla and increases its osmotic gradient. This gradient helps the collecting duct to reabsorb water and produce concentrated urine.

7. Functions of kidney

The kidney performs several vital functions for homeostasis. It excretes nitrogenous wastes such as urea, excess salts and drugs in urine, regulates the water and osmotic balance of blood, and maintains acid-base balance by secreting H+ and reabsorbing bicarbonate. It also regulates blood pressure through renin and fluid volume control, secretes erythropoietin for red blood cell formation, and activates vitamin D for calcium metabolism.

- **Excretion:** excretes nitrogenous wastes such as urea, excess salts and drugs in urine.
- **Water and osmotic balance:** regulates the water and osmotic balance of blood.
- **Acid-base balance:** maintains acid-base balance by secreting H+ and reabsorbing bicarbonate.
- **Blood pressure:** regulates blood pressure through renin and fluid volume control.
- **Red blood cell formation:** secretes erythropoietin for red blood cell formation.
- **Calcium metabolism:** activates vitamin D for calcium metabolism.

8. Kidney as an endocrine organ

Besides excretion, the kidney secretes important hormones, so it also acts as an endocrine organ. It produces erythropoietin, which stimulates red blood cell formation in bone marrow, and renin, which helps regulate blood pressure and salt balance. It also activates vitamin D, which is needed for calcium absorption.

- **Erythropoietin:** an important hormone necessary for the synthesis of RBC's, produced by the renal cortex. It is released by the kidney, mainly from the renal cortex, when oxygen in blood falls. It stimulates the bone marrow to produce more red blood cells.
- **Renin:** helps regulate blood pressure and salt balance.
- **Vitamin D:** the kidney activates vitamin D, which is needed for calcium absorption.

Kidney Disorders, Failure and Transplant

LESSON 5

1. How kidney stones are formed

Urine contains dissolved minerals and salts such as calcium oxalate, calcium phosphate and uric acid. When urine becomes too concentrated, or these substances increase, they crystallize and gradually deposit in the kidney or urinary tract to form stones. Low water intake, excess salts in diet and urinary infections increase the risk of stone formation.

2. Lithonephritis

Lithonephritis is the inflammation in kidneys due to irritation of kidney stones. It is the inflammation of the kidney caused by the irritation of kidney stones.

- **Edema:** swelling due to fluid.
- **Sarcoma:** a cancer.
- **Encephalitis:** inflammation of the brain.

3. Impact if kidneys lose filtering ability

When kidneys lose their filtering ability, urea and other wastes accumulate in the blood and water and salt balance is disturbed. This causes nausea, swelling, high blood pressure, anaemia, weakness and disturbance of acid-base balance. In complete kidney failure the condition becomes life-threatening unless dialysis or transplant is provided.

4. Causes of kidney failure

Kidney failure may be sudden or gradual and is caused by conditions such as uncontrolled diabetes and high blood pressure, glomerular inflammation, kidney stones, urinary tract obstruction, infections and toxic drugs that damage the nephrons. As filtration fails, wastes and fluids accumulate in the body and disturb blood chemistry.

5. Treatment of kidney failure

Treatment includes controlling the cause and diet, haemodialysis or peritoneal dialysis and continuous renal replacement therapy to artificially filter the blood, and in severe permanent failure, kidney transplant from a suitable donor.

6. Problems associated with kidney transplant

The main problem is rejection, because the immune system of the recipient recognizes the donor kidney as foreign and attacks it, so immunosuppressive drugs are needed for life. These drugs increase the risk of infections and other side effects. Other problems include shortage of suitable donors, the need for close tissue matching, high cost and possible failure of the transplanted kidney after some years.

- **Rejection:** the immune system of the recipient recognizes the donor kidney as foreign and attacks it.
- **Immunosuppressive drugs:** needed for life; they increase the risk of infections and other side effects.
- **Other problems:** shortage of suitable donors, the need for close tissue matching, high cost and possible failure of the transplanted kidney after some years.

Thermoregulation

LESSON 6

1. Thermoregulation

Thermoregulation is the regulation of body temperature as part of homeostasis. Body temperature must stay within a suitable limit, and it rises or falls when heat production and heat loss are disturbed.

2. Hyperthermia

If the surrounding environment becomes hot or a person does strenuous physical activity, the heat produced inside the body raises body temperature above suitable limit; this condition is called hyperthermia. Hyperthermia is the rise of body temperature above the normal suitable limit due to external heat or heavy physical activity.

3. Hypothermia

Hypothermia is the opposite condition to hyperthermia, in which body temperature falls below normal. It is the condition in which body temperature falls below the normal limit, usually due to prolonged exposure to cold.

Condition	Body temperature	Cause
Hyperthermia	Rises above the normal suitable limit	External heat or heavy physical activity
Hypothermia	Falls below the normal limit	Usually prolonged exposure to cold

4. How body overcomes hypothermia

The body overcomes hypothermia by shivering, which produces heat through rapid muscle contractions, and by constriction of skin blood vessels, which reduces heat loss. Behavioural responses such as seeking shelter and wearing warm clothes also help restore normal temperature.

- **Shivering:** produces heat through rapid muscle contractions.
- **Constriction of skin blood vessels:** reduces heat loss.
- **Behavioural responses:** seeking shelter and wearing warm clothes help restore normal temperature.

Support and Movement

CHAPTER 2

Bone Cells, Bone Formation and Cartilage

LESSON 7

1. Bone cells

Bone is built, maintained and remodelled by different types of bone cells.

- **Osteoclasts:** large cells that break down and resorb old bone tissue during growth and remodeling; they tear down bone during the building and remodeling process.
- **Osteoblasts:** do the opposite work of osteoclasts and build new bone.
- **Osteocytes:** maintain the mature bone.

2. Periosteal bone collar

During bone formation, as the periosteum calcifies, it gives rise to a thin plate of compact bone called the periosteal bone collar. The periosteum around the shaft calcifies and forms this collar. This collar supports the developing bone and is later thickened as ossification proceeds.

3. Cartilage and perichondrium

Cartilage is a soft flexible form of connective tissue. It is surrounded by a layer called the perichondrium.

- **Perichondrium:** a dense connective tissue layer that covers cartilage externally, nourishes it and helps in its growth and repair.
- **Periosteum:** the corresponding covering layer of bone, not of cartilage.

4. How cartilage is useful for the skeletal system

Cartilage covers the ends of bones at joints, where it reduces friction and absorbs shocks during movement. It also gives shape and flexible support to structures such as the nose, ears and trachea, and in children it forms the model from which bones develop and lengthen. Being soft and flexible, it protects the joints from wear and damage.

5. Different types of cartilage

- **Hyaline cartilage:** glassy and smooth; covers the ends of bones at joints and is found in the nose, trachea and ribs, where it provides support with flexibility.
- **Elastic cartilage:** contains many elastic fibers, is highly flexible, and is found in the external ear and epiglottis.
- **Fibrocartilage:** contains dense collagen fibers, is the strongest type, and is found in intervertebral discs and the knee, where it resists compression and absorbs shocks.

Type of cartilage	Features and places
Hyaline cartilage	Glassy and smooth; ends of bones at joints, nose, trachea and ribs; support with flexibility
Elastic cartilage	Many elastic fibers, highly flexible; external ear and epiglottis
Fibrocartilage	Dense collagen fibers, strongest type; intervertebral discs and knee; resists compression and absorbs shocks

The Human Skeleton — Pelvic Girdle and Joints

LESSON 8

1. The pelvic girdle

The ring like shape of the girdle is due to the joining of the coxal bone with the sacrum of the vertebral column. The two coxal bones join the sacrum behind and meet each other in front at the pubic symphysis, a cartilaginous joint, in between the pubic part. This arrangement completes the ring-like pelvic girdle that supports the trunk and protects the pelvic organs.

- **Coxal bones:** join the sacrum behind.
- **Pubic symphysis:** the joint in front where the two coxal bones meet each other.
- **Role of the pelvic girdle:** supports the trunk and protects the pelvic organs.

2. Fibrous joints

Fibrous joints hold the bone by dense connective tissue containing collagenous fibers. In fibrous joints the bones are held together directly by dense connective tissue rich in collagen fibers, allowing little or no movement. Synovial joints, in contrast, have a fluid-filled cavity and allow free movement.

3. Comparison of major types of joints

Joints differ in how the bones are joined and in how much movement they allow.

Type of joint	How bones are joined	Movement	Examples
Fibrous joints	Joined by dense connective tissue	No or very little movement	Between skull bones
Cartilaginous joints	Joined by cartilage	Slight movement	Between vertebrae and at the pubic symphysis
Synovial joints	Have a fluid-filled synovial cavity	Free movement in one or more directions	Knee, shoulder and elbow joints

Fracture Repair, Sprains and First Aid for Joint Injuries

LESSON 9

1. How the healing of a fractured bone is initiated

When a bone fractures, blood vessels in the bone and periosteum are torn, causing pain and bleeding that forms a clot called a hematoma around the broken ends. The pain and swelling signal injury, and the hematoma provides the framework in which repair begins. In the next steps a cartilaginous callus joins the broken ends, which is then replaced by bone and finally remodeled.

2. Repair process of simple fractures

Repair of a simple fracture takes place in a fixed sequence of steps.

- **Hematoma formation:** bleeding from torn vessels forms a hematoma (blood clot) around the broken ends, which seals the area.
- **Callus formation:** an internal and external callus of cartilage forms and joins the two sides temporarily.
- **Replacement by bone:** the cartilage of the callus is replaced by trabecular (spongy) bone, making the union firm.
- **Bone remodeling stage:** osteoclasts remove dead and excess bone and osteoblasts lay down compact bone, restoring the original shape and strength.

In the bone remodeling stage, old bone tissues are removed and replaced with new ones that join both fractured sides. Osteoclasts remove old and dead bone and osteoblasts lay down new bone that firmly joins both fractured sides, and the bone is gradually remodeled to its original shape and strength.

3. Sprains

A sprain is related to the mild to severe grade of stretching or tearing of a ligament that holds the bones of a joint together. Spasm, tetany and cramps are abnormal conditions of muscles, not of ligaments.

4. First aid treatment for joint injuries and fractures

- **Keep the part still:** the injured part should be kept still and supported, for example with a splint or sling, so that the broken bone or injured joint does not move and cause further damage.
- **Cold packs:** can be applied to reduce pain and swelling.
- **Raised limb:** the limb may be kept raised.
- **Bandaging:** tight bandaging should be avoided.
- **Hospital treatment:** the patient should then be taken to hospital promptly for X-ray, proper alignment and plastering or other treatment.

Types of Muscles and Structure of Skeletal Muscle

LESSON 10

1. Comparison of types of muscles

Type of muscle	Features
Skeletal muscles	Attached to bones, striated and voluntary; produce body movements such as walking and lifting
Smooth muscles	Found in the walls of involuntary organs like the gut and blood vessels, non-striated and work automatically
Cardiac muscle	Found only in the heart, striated but involuntary; contracts rhythmically throughout life without fatigue

2. Cardiac muscle and syncytium

The heart muscles are arranged in a branching network where cells are joined together and make a syncytium. Cardiac muscle cells branch and are joined end to end by intercalated discs, so the whole tissue works as a single functional unit called a syncytium. This allows the heart to contract in a coordinated wave.

3. Structure of skeletal muscles

A skeletal muscle is made of bundles of long muscle fibers held together by connective tissue layers.

- **Muscle fiber:** each individual fiber is covered by the endomysium, and its cell membrane is the sarcolemma.
- **Fascicle:** muscle fibers are gathered in many groups; each group of fibers is called a fascicle, which is surrounded by a membrane called the perimysium.
- **Whole muscle:** covered by the epimysium.

Layer / membrane	What it covers
Endomysium	Individual fibers
Perimysium	Each fascicle, a bundle of muscle fibers
Epimysium	The whole muscle
Sarcolemma	The cell membrane of a fiber

4. Ultra structure of skeletal muscle fiber

Under the sarcolemma, each fiber contains many rod-like myofibrils made of repeating units called sarcomeres, which are the units of contraction. Each sarcomere has thick myosin filaments in the A band and thin actin filaments in the I band, separated by Z lines, and their sliding over each other shortens the fiber and contracts the muscle.

- **Myofibrils:** rod-like structures inside each fiber.
- **Sarcomere:** the repeating unit of a myofibril and the unit of contraction.
- **A band:** contains thick myosin filaments.
- **I band:** contains thin actin filaments.
- **Z lines:** separate the sarcomeres.

Muscle Contraction — Antagonistic Action, Calcium and the Cross-Bridge Cycle

LESSON 11

1. Antagonistic muscles

Muscles work in antagonistic pairs, so when one muscle, the agonist, contracts to produce a movement, the opposite antagonist muscle relaxes. An antagonist muscle is relaxed while the muscle opposite to it is contracted. For example, during flexion of the forearm the biceps contracts while the triceps relaxes.

2. Why calcium is essential for cross bridge formation

When a muscle is stimulated, calcium ions are released from the sarcoplasmic reticulum into the muscle fiber. Calcium binds with troponin, which moves tropomyosin away from the active sites on actin, so the myosin heads can attach to actin and form cross bridges. Without calcium the active sites remain covered and contraction cannot begin.

3. How nervous coordination regulates muscle contraction

A motor neuron carries the impulse to the muscle, and at the neuromuscular junction it releases acetylcholine, which generates an action potential in the muscle fiber membrane. This action potential spreads through the T-tubules and causes the sarcoplasmic reticulum to release calcium ions, which trigger cross bridge formation and contraction. When the nerve impulse stops, calcium is pumped back and the muscle relaxes, so the nervous system controls both the start and the strength of contraction.

- **Motor neuron:** carries the impulse to the muscle.
- **Acetylcholine:** released at the neuromuscular junction; generates an action potential in the muscle fiber membrane.
- **T-tubules:** spread the action potential through the fiber.
- **Sarcoplasmic reticulum:** releases calcium ions, which trigger cross bridge formation and contraction.

4. Cross-bridge cycle of muscle contraction

In the cross-bridge cycle, calcium first exposes the active sites on actin, and the energized myosin head, carrying ADP and phosphate, attaches to actin to form a cross bridge. The myosin head then bends in a power stroke, pulling the actin filament towards the center of the sarcomere and releasing ADP and phosphate. A new ATP molecule binds to the myosin head, which detaches it from actin, and breakdown of this ATP re-energizes the head so the cycle can repeat. Repeated cycles slide the actin filaments over myosin and shorten the sarcomere, producing contraction.

- **Cross bridge formation:** the energized myosin head, carrying ADP and phosphate, attaches to actin.
- **Power stroke:** the myosin head bends, pulling the actin filament towards the center of the sarcomere and releasing ADP and phosphate.
- **Detachment:** a new ATP molecule binds to the myosin head and detaches it from actin.
- **Re-energizing:** breakdown of this ATP re-energizes the head so the cycle can repeat.

Muscle Disorders — Tetany and Tetanus

LESSON 12

1. Tetany

Tetany is a condition of sustained, painful muscle spasms caused by low calcium in the blood, often due to parathyroid deficiency, and it affects mainly the hands, feet and face. Tetany is a metabolic disorder.

2. Tetanus

Tetanus is an infectious disease caused by the bacterium Clostridium tetani entering a wound, whose toxin causes severe, continuous contraction of muscles including lockjaw. Tetanus is a bacterial infection.

3. How tetany is different from tetanus

Feature	Tetany	Tetanus
Nature	A metabolic disorder	A bacterial infection
Cause	Low calcium in the blood, often due to parathyroid deficiency	The bacterium Clostridium tetani entering a wound; its toxin acts on the muscles
Effect	Sustained, painful muscle spasms, mainly in the hands, feet and face	Severe, continuous contraction of muscles including lockjaw

Nervous Coordination

CHAPTER 3

The Neuron — Structure, Types and Function

LESSON 13

1. Components of the nervous system

The nervous system makes rapid communication and coordination in the body possible. Neurons generate and conduct nerve impulses, so sensory neurons carry information from receptors to the CNS, interneurons integrate it, and motor neurons carry commands to effectors.

2. Structure of a neuron

A neuron consists of three main parts: the cell body (soma), the dendrites and the axon.

- **Cell body (soma):** contains the nucleus and cytoplasm and controls the metabolic activities of the cell.
- **Dendrites:** short branched processes that receive impulses and carry them towards the cell body.
- **Axon:** a long fiber that carries impulses away from the cell body to other neurons, muscles or glands, often covered by a myelin sheath with nodes of Ranvier.

3. Types of neurons by function

- **Sensory neurons:** carry information from receptors to the CNS.
- **Interneurons:** integrate information within the CNS.
- **Motor neurons:** carry commands to effectors.

Most neurons of the brain are interneurons, which lie entirely within the central nervous system and connect sensory and motor neurons. They are responsible for integration, processing and decision making in the brain and spinal cord.

4. Why interneurons are called association neurons

Interneurons lie between sensory and motor neurons within the central nervous system and connect, or associate, them with each other. They receive information from sensory neurons, process and integrate it, and pass suitable commands to motor neurons. Because of this associating and connecting role, they are called association neurons.

5. How the nervous system alters target cells

A nervous system can alter activities in its target cells in muscles and glands because the target cells have receptor proteins for the signals released by the nervous system. Neurons release neurotransmitters at their endings, and only those muscle or gland cells that carry matching receptor proteins respond to them. This receptor specificity allows the nervous system to alter the activity of selected target cells.

Type of neuron	Main role
Sensory neurons	Carry information from receptors to the CNS
Interneurons	Integration, processing and decision making in the brain and spinal cord; connect sensory and motor neurons
Motor neurons	Carry commands to effectors

Nerve Impulse — Polarized Membrane, Threshold and Action Potential

LESSON 14

1. Why the neurolemma is polarized

At rest, the inside of the neuron contains more potassium ions and negatively charged proteins, while the outside has more sodium ions, and the sodium-potassium pump keeps pumping Na+ out and K+ in. As a result the inner surface of the neurolemma (axon membrane) is negative and the outer surface is positive, with a resting potential of about -70 mV. This separation of charge across the membrane is why the neurolemma is called polarized.

2. Hyperpolarization

For a neuron with an initial membrane potential at -70mv, an increase in the movement of potassium ions out of that neuron's cytoplasm would result in hyperpolarization of the neuron. Potassium ions carry positive charge, so their increased movement out of the cell makes the inside even more negative than -70 mV. This increase in negativity is called hyperpolarization, and it makes the neuron less likely to fire.

3. Threshold potential

Threshold potential is the minimum level of depolarization, about -55 mV, which a stimulus must produce in the neuron membrane for an action potential to be generated. A stimulus weaker than threshold produces no impulse, while any stimulus reaching threshold produces a full action potential. It therefore follows the all-or-none principle.

- **Resting potential:** about -70 mV.
- **Threshold potential:** about -55 mV.
- **All-or-none principle:** a stimulus weaker than threshold produces no impulse, while any stimulus reaching threshold produces a full action potential.

4. Mechanism of action potential

The action potential is a rapid, temporary reversal of membrane potential produced when a stimulus reaches threshold. First, voltage-gated sodium channels open and Na+ rushes into the axon, so the inside becomes positive (up to about +40 mV); this is depolarization. Then sodium channels close and potassium channels open, K+ moves out, and the membrane returns to its negative resting value; this is repolarization, sometimes followed by brief hyperpolarization. The sodium-potassium pump then restores the original ion distribution, and the action potential travels along the axon as each neighboring region is depolarized in turn.

Stage	What happens
Depolarization	Voltage-gated sodium channels open and Na+ rushes into the axon, so the inside becomes positive (up to about +40 mV)
Repolarization	Sodium channels close and potassium channels open, K+ moves out, and the membrane returns to its negative resting value, sometimes followed by brief hyperpolarization
Restoration	The sodium-potassium pump restores the original ion distribution

Conduction in Myelinated Axons and Synaptic Transmission

LESSON IS

1. Why impulses move faster in myelinated neurons

Action potentials move along axons more rapidly in myelinated than in non-myelinated axons. In myelinated neurons the myelin sheath insulates the axon, and the action potential is generated only at the exposed nodes of Ranvier. The impulse therefore jumps rapidly from node to node instead of traveling continuously along the whole membrane. Larger diameter also increases speed.

2. Saltatory conduction

Saltatory conduction is the jumping movement of the nerve impulse from one node of Ranvier to the next along a myelinated axon. The action potential is regenerated only at the nodes, while the myelinated parts are insulated. This makes conduction fast and also saves energy for the neuron. This saltatory conduction makes conduction many times faster than in non-myelinated neurons.

3. The synapse — presynaptic and postsynaptic membranes

The surface on a neuron that discharges vesicles is the presynaptic membrane. The presynaptic membrane of the axon terminal discharges synaptic vesicles containing neurotransmitter into the synaptic cleft. The postsynaptic membrane only receives the transmitter through its receptors.

Neurotransmitter receptors are located on the postsynaptic membrane, facing the synaptic cleft. When transmitter released from the presynaptic side binds to them, ion channels open and a postsynaptic potential is produced.

- **Presynaptic membrane:** discharges synaptic vesicles containing neurotransmitter into the synaptic cleft.
- **Postsynaptic membrane:** receives the transmitter through its receptors, which are located on it.

4. Electrical and chemical synapse

Electrical synapse	Chemical synapse
The impulse passes directly through gap junctions between cells, so transmission is very fast and can be two-way	A neurotransmitter carries the signal across the synaptic cleft, so transmission is slower and one-way

5. Excitatory and inhibitory neurotransmitters

An inhibitory postsynaptic potential is produced when the membrane becomes more permeable to potassium ions (and chloride ions), so positive charge leaves the cell and the inside becomes more negative. This hyperpolarization moves the membrane away from threshold and inhibits firing. An inhibitory postsynaptic potential occurs in a membrane made more permeable to potassium ions.

- **Excitatory neurotransmitters:** such as acetylcholine and glutamate depolarize the postsynaptic membrane and promote firing.
- **Inhibitory neurotransmitters:** such as GABA hyperpolarize it and prevent firing.
- **GABA:** GABA (gamma-aminobutyric acid) is the main inhibitory neurotransmitter of the brain, reducing the activity of neurons and preventing over-excitation. It is the major inhibitory neurotransmitter of the brain.
- **Acetylcholine and epinephrine:** mainly act as excitatory transmitters at most of their sites.
- **Endorphins:** natural pain-relieving chemicals produced by the brain that block pain pathways and produce a feeling of well-being.

Feature	Excitatory neurotransmitters	Inhibitory neurotransmitters
Examples	Acetylcholine and glutamate	GABA
Effect on postsynaptic membrane	Depolarize the postsynaptic membrane and promote firing	Hyperpolarize it and prevent firing

Central Nervous System — Anatomy and Physiology of the Human Brain

LESSON 16

1. Protection and divisions of the brain

The human brain is protected inside the skull by the bony cranium, three meninges and cerebrospinal fluid, and is divided into forebrain, midbrain and hindbrain.

2. Forebrain

The forebrain includes the cerebrum, the largest part, whose cerebral cortex controls intelligence, memory, voluntary actions and sensations, and the diencephalon (thalamus and hypothalamus), which relays sensory information and regulates hunger, thirst, body temperature and endocrine activity.

- **Cerebrum:** the largest part of the forebrain; its cerebral cortex controls intelligence, memory, voluntary actions and sensations.
- **Diencephalon:** part of the forebrain; the thalamus and hypothalamus relay sensory information and regulate hunger, thirst, body temperature and endocrine activity.

3. Midbrain

The small midbrain acts mainly as a relay center for visual and auditory information. The midbrain forms the upper part of the brainstem.

4. Hindbrain

The hindbrain consists of the cerebellum, which coordinates movement and balance, and the pons and medulla, which control vital involuntary functions such as breathing, heartbeat and blood pressure.

- **Cerebellum:** part of the hindbrain, not of the midbrain; it coordinates movement and balance.
- **Pons and medulla:** part of the hindbrain; they control vital involuntary functions such as breathing, heartbeat and blood pressure.

Division	Parts	Main functions
Forebrain	Cerebrum and diencephalon (thalamus and hypothalamus)	Intelligence, memory, voluntary actions and sensations; relay of sensory information and regulation of hunger, thirst, body temperature and endocrine activity
Midbrain	Upper part of the brainstem	Relay center for visual and auditory information
Hindbrain	Cerebellum, pons and medulla	Coordination of movement and balance; vital involuntary functions such as breathing, heartbeat and blood pressure

Sensory Receptors and the Reflex Arc

LESSON 17

1. Why receptors are called transducers

A transducer is a device that converts one form of energy into another form. Sensory receptors convert the energy of a stimulus, such as light, sound, pressure or heat, into electrical energy in the form of nerve impulses. Because of this conversion of stimulus energy into action potentials, receptors are called biological transducers.

2. Sensory receptors and their working

Sensory receptors are specialized cells or neuron endings that detect a specific stimulus and convert it into nerve impulses. When a stimulus of sufficient strength acts on a receptor, it produces a generator potential, and if this reaches threshold, action potentials are fired whose frequency reflects the strength of the stimulus. These impulses travel through sensory neurons to the CNS, where the sensation is perceived and a suitable response is organized.

3. Types of receptors

They are classified by the stimulus they detect.

- **Mechanoreceptors:** for touch, pressure and sound; mechanoreceptors detect sound vibrations.
- **Chemoreceptors:** for taste, smell and blood chemicals; chemoreceptors detect solute concentration.
- **Thermoreceptors:** for temperature; thermoreceptors detect heat.
- **Photoreceptors:** (rods and cones) for light.
- **Nociceptors:** for pain; pain receptors that respond to damaging stimuli and produce the sensation of pain.
- **Pacinian corpuscles:** pressure receptors deep in the skin and tissues that detect deep pressure and vibrations.

Pain receptors (nociceptors) respond to damaging mechanical, thermal or chemical stimuli, not to electricity as their specific energy.

Receptor	Stimulus / energy it detects
Mechanoreceptors	Touch, pressure and sound
Chemoreceptors	Taste, smell and blood chemicals; solute concentration
Thermoreceptors	Temperature; heat
Photoreceptors (rods and cones)	Light
Nociceptors	Damaging mechanical, thermal or chemical stimuli; pain
Pacinian corpuscles	Deep pressure and vibrations

4. Reflex arc

A reflex arc is the pathway of a reflex action: receptor, sensory neuron, interneuron in the CNS, motor neuron and effector.

- **Receptor:** detects the stimulus.
- **Sensory neuron:** carries the impulse from the receptor.
- **Interneuron in the CNS:** connects the sensory and motor neurons.
- **Motor neuron:** carries the command from the CNS.
- **Effector:** carries out the response.

Peripheral Nervous System, Autonomic Nervous System and Disorders of the CNS

LESSON 18

1. CNS and PNS

CNS	PNS
Consists of the brain and spinal cord and integrates information	Consists of cranial and spinal nerves and ganglia and carries impulses between the CNS and the body

2. Sympathetic and parasympathetic nervous system

The sympathetic system prepares the body for emergency (fight or flight), increasing heartbeat and dilating pupils, while the parasympathetic system works during rest, slowing the heart and promoting digestion; the two act antagonistically to maintain balance.

Sympathetic nervous system	Parasympathetic nervous system
Prepares the body for emergency (fight or flight), increasing heartbeat and dilating pupils	Works during rest, slowing the heart and promoting digestion

3. Vascular disorders of the CNS

Vascular disorders of the CNS are caused by disturbed blood supply to the brain, the most important being stroke, in which a clot blocks a cerebral vessel or a vessel bursts and bleeds, killing the brain tissue deprived of oxygen and causing paralysis, loss of speech or death.

- **Stroke:** the death of brain tissue due to blockage or bursting of a blood vessel supplying the brain.

4. Infectious disorders of the CNS

Infectious disorders are caused by bacteria, viruses or other pathogens that invade the CNS, such as meningitis, the inflammation of the meninges, and encephalitis, the inflammation of the brain itself. These infections cause fever, severe headache, neck stiffness and damage to nervous tissue, and need prompt medical treatment.

- **Meningitis:** the inflammation of the meninges, the protective membranes around the brain and spinal cord, usually caused by bacterial or viral infection.
- **Encephalitis:** the inflammation of the brain itself.

Disorder	Cause / nature
Stroke	Death of brain tissue due to blockage or bursting of a blood vessel supplying the brain; a clot blocks a cerebral vessel or a vessel bursts and bleeds, killing the brain tissue deprived of oxygen
Meningitis	Inflammation of the meninges, usually caused by bacterial or viral infection
Encephalitis	Inflammation of the brain itself, caused by bacteria, viruses or other pathogens that invade the CNS

Chemical Coordination

CHAPTER 4

Hormones and Their Mode of Action

LESSON 19

1. Hormones — the chemical messengers

Hormones are chemical messengers secreted by endocrine glands directly into the blood, which regulate the activity of specific target organs. Hormones are chemical substances produced by endocrine glands in very small amounts and carried by the blood to distant target organs. Like messengers, they carry information and instructions from one part of the body to another and tell the target cells to start, stop or adjust a particular activity. For this reason they are called chemical messengers.

Hormones are chemical messengers that coordinate activities within the same organism, being carried by the blood from endocrine glands to their target organs. Communication between different organisms is done by pheromones, not by hormones.

- **Produced by:** endocrine glands.
- **Chemical nature:** modified amino acids, peptides or steroid molecules.
- **Carried by:** the circulatory system.
- **Action:** within the same organism, on specific target organs.

2. Hormones formed from tyrosine

Epinephrine and the other catecholamines of the adrenal medulla are derived from the amino acid tyrosine, and thyroxine of the thyroid gland is also formed from tyrosine units. Hence epinephrine, catecholamines and thyroxin all share tyrosine as their starting material.

3. Definition of a hormone

A hormone is a chemical messenger produced by an endocrine gland, released directly into the blood in small quantity, and carried to a specific target organ where it regulates its activity. In both modes of action the hormone is recognized only by cells carrying its specific receptor, which is why its action remains target-specific.

4. Action of hydrophilic hormones

Hydrophilic (water-soluble) hormones such as peptides and catecholamines cannot pass through the lipid plasma membrane of the target cell. They therefore bind to receptors on the cell surface, and the message is carried inside the cell by a second messenger such as cyclic AMP, which activates enzymes and produces the response. The hormone itself acts as the first messenger outside the cell.

Hydrophilic hormones cannot cross the cell membrane, so they bind to surface receptors and act through a second messenger such as cyclic AMP, which activates enzymes inside the cell and produces a rapid response.

5. Why hydrophilic hormones need secondary messengers

Hydrophilic hormones need secondary messengers because they cannot pass through the lipid plasma membrane of the target cell. They bind to receptors on the cell surface, and the message is carried inside the cell by a second messenger such as cyclic AMP, which activates enzymes and produces the response.

6. Action of lipophilic hormones

Lipophilic hormones such as steroids and thyroxine pass through the membrane, bind to receptors inside the cell or nucleus, and switch on specific genes, producing a slower but longer-lasting response through new protein synthesis.

Thyroid and steroid hormones are lipid soluble, so they pass through the plasma membrane and bind to receptors inside the cell, in the cytoplasm or nucleus. The hormone-receptor complex then switches specific genes on or off, changing protein synthesis in the target cell.

7. Lipophilic and hydrophilic hormones

Feature	Lipophilic hormones	Hydrophilic hormones
Solubility	Lipid soluble (steroids, thyroxine)	Water soluble (peptides, catecholamines)
Passage through membrane	Pass through the cell membrane	Cannot cross the membrane
Receptors	Bind to internal receptors	Act through surface receptors and second messengers

Pituitary Gland

LESSON 20

1. The pituitary gland

The pituitary gland is a pea-sized master gland hanging from the hypothalamus below the brain, and it has two main lobes. Because it controls so many other glands through its tropic hormones, the pituitary is called the master endocrine gland.

- **Anterior lobe (adenohypophysis):** glandular in origin.
- **Posterior lobe (neurohypophysis):** nervous in origin.

2. Anterior pituitary hormones

The anterior pituitary secretes six main hormones: growth hormone (GH or somatotropin) for body growth, thyroid stimulating hormone (TSH) for the thyroid, adrenocorticotropic hormone (ACTH) for the adrenal cortex, follicle stimulating hormone (FSH) and luteinizing hormone (LH) for the gonads, and prolactin for milk production in the mammary glands. Most of these, except GH and prolactin, are tropic hormones that control other endocrine glands.

The anterior lobe (adenohypophysis) is glandular and secretes growth hormone, prolactin and the tropic hormones TSH, ACTH, FSH and LH, which control growth, milk production and the thyroid, adrenal cortex and gonads respectively.

Hormone	Main action
Growth hormone (GH or somatotropin)	Body growth
Thyroid stimulating hormone (TSH)	Controls the thyroid
Adrenocorticotropic hormone (ACTH)	Controls the adrenal cortex
Follicle stimulating hormone (FSH)	Acts on the gonads
Luteinizing hormone (LH)	Acts on the gonads
Prolactin	Milk production in the mammary glands

3. Tropic hormones and thyroxine

The anterior pituitary secretes thyroid stimulating hormone (TSH), a tropic hormone, which directly stimulates the thyroid gland to release thyroxine. Parathyroid hormone, calcitonin and epinephrine are not controlled by pituitary tropic hormones in this way.

4. Posterior pituitary — neurohypophysis

The posterior pituitary is made of nervous tissue and is actually an extension of the hypothalamus, connected to it by nerve fibers. It does not manufacture its own hormones; it only stores and releases oxytocin and ADH made by hypothalamic neurons. Because of this nervous origin and structure, it is called the neurohypophysis, unlike the glandular anterior pituitary (adenohypophysis).

The posterior lobe (neurohypophysis) is nervous in origin; it stores and releases oxytocin, which contracts the uterus and ejects milk, and ADH, which promotes reabsorption of water in the kidneys.

5. Oxytocin and ADH

Oxytocin and ADH (vasopressin) are actually synthesized by neurosecretory cells of the hypothalamus. They are then transported down the axons and stored in the neurohypophysis (posterior pituitary), from where they are released into the blood.

- **Oxytocin:** strongly stimulates contraction of the smooth muscle of the uterus during childbirth. It also causes contraction of mammary glands for the ejection of milk after birth.
- **ADH:** promotes reabsorption of water in the kidneys.

Oxytocin, released from the posterior pituitary, stimulates the contraction of uterine muscle.

Thyroid and Parathyroid Glands

LESSON 21

1. Thyroid gland and thyroxine

The thyroid gland releases thyroxine under the control of thyroid stimulating hormone (TSH) from the anterior pituitary. Thyroxin is also formed from tyrosine units.

2. Calcitonin and parathyroid hormone

Parathyroid hormone (PTH) raises blood calcium by releasing calcium from bones, while calcitonin from the thyroid lowers blood calcium by promoting its deposition in bones. The two hormones thus act antagonistically to keep blood calcium constant.

Calcitonin, from the thyroid, lowers blood calcium by depositing it in bones, while parathyroid hormone raises blood calcium by releasing it from bones and increasing its absorption.

Feature	Calcitonin	Parathyroid hormone (PTH)
Source	Thyroid	Parathyroid
Effect on blood calcium	Lowers blood calcium by depositing it in bones	Raises blood calcium by releasing it from bones and increasing its absorption
Relation	Antagonistic to PTH	Antagonistic to calcitonin

3. Hypothyroidism and hyperthyroidism

Hypothyroidism	Hyperthyroidism
Under-secretion of thyroxine	Over-secretion of thyroxine
Low metabolic rate, weight gain and sluggishness	High metabolic rate, weight loss, rapid heartbeat and nervousness
Cretinism in children, myxedema in adults	—

4. Goiter

Goiter is the enlargement of the thyroid gland, usually caused by iodine deficiency, seen as a swelling in the neck. Goiter is the enlargement of the thyroid gland, usually due to iodine deficiency, and has nothing to do with the adrenal medulla.

Pancreas and Adrenal Glands

LESSON 22

1. Pancreas — a mixed gland

The pancreas works in both ways, so it is a mixed gland. Its acinar cells act as an exocrine part and pour pancreatic juice containing digestive enzymes through a duct into the duodenum. Its islets of Langerhans act as the endocrine part and release hormones directly into the blood: insulin from beta cells, glucagon from alpha cells and somatostatin from delta cells.

2. Insulin and glucagon

Insulin	Glucagon
From beta cells of the pancreas	From alpha cells of the pancreas
Lowers blood glucose by promoting its uptake and glycogen formation	Raises blood glucose by breaking down glycogen in the liver

3. Adrenal glands — position

There are two adrenal (suprarenal) glands, one sitting like a cap on top of each kidney. Each gland has an outer cortex and an inner medulla with different functions.

4. Adrenal cortex and corticosteroids

All hormones of the adrenal cortex are chemically steroids and are synthesized from cholesterol. Because they are steroid hormones produced by the cortex, they are collectively called corticosteroids. They include glucocorticoids (e.g. cortisol), mineralocorticoids (e.g. aldosterone) and small amounts of sex steroids.

The cortex secretes corticosteroids - glucocorticoids such as cortisol, which regulate carbohydrate metabolism and stress responses, mineralocorticoids such as aldosterone, which regulate salt, water and blood pressure, and small amounts of sex hormones.

The adrenal cortex produces steroid hormones, namely glucocorticoids, mineralocorticoids and sex steroids, all made from cholesterol. Removal of the cortex would therefore directly abolish this steroid group, while peptide and amino-acid-derived hormones are made elsewhere.

5. Glucocorticoids and stress

Glucocorticoids such as cortisol are secreted by the adrenal cortex during stress and promote gluconeogenesis, the synthesis of glucose from amino acids and other non-carbohydrate substrates. This raises blood glucose to meet the increased energy demand of the stress condition. Glucocorticoids are secreted by the adrenal gland in response to stress and promote the synthesis of glucose from non-carbohydrate substrates.

6. Adrenal medulla

The medulla secretes the catecholamines epinephrine and norepinephrine, which prepare the body for emergency by increasing heartbeat, blood pressure and blood glucose. Together the two parts help the body handle stress and maintain salt, water and energy balance.

Part of adrenal gland	Hormones	Function
Cortex — glucocorticoids (e.g. cortisol)	Corticosteroids	Regulate carbohydrate metabolism and stress responses; promote gluconeogenesis
Cortex — mineralocorticoids (e.g. aldosterone)	Corticosteroids	Regulate salt, water and blood pressure
Cortex — sex steroids	Small amounts of sex hormones	—
Medulla	Epinephrine and norepinephrine (catecholamines)	Prepare the body for emergency by increasing heartbeat, blood pressure and blood glucose

Gonads and Other Hormones — GI, Placental, Kidney and Other Tissues

LESSON 23

1. Gonadotropic hormones

GnTH (gonadotropic hormones) are FSH and LH from the anterior pituitary that stimulate the gonads to produce gametes and sex hormones.

- **FSH (follicle stimulating hormone):** a gonadotropic hormone acting on the gonads.
- **LH (luteinizing hormone):** a gonadotropic hormone acting on the gonads.

2. Hormones of other endocrine tissues

Besides the main glands, several other tissues produce hormones with important functions.

3. Gastrointestinal tract

The gastrointestinal tract secretes gastrin, secretin and cholecystokinin, which coordinate digestion by controlling gastric juice, pancreatic juice and bile release.

4. Kidneys

The kidneys secrete erythropoietin for red blood cell formation and renin for blood pressure regulation.

- **Erythropoietin:** a hormone from the kidneys that stimulates the bone marrow to produce red blood cells when blood oxygen is low.
- **Renin:** secreted by the kidneys for blood pressure regulation.

5. Heart

The heart releases atrial natriuretic hormone, which lowers blood pressure by increasing salt and water excretion. Atrial natriuretic hormone is released by the heart atria when blood pressure/volume rises, and it increases excretion of sodium and water by the kidneys to lower blood pressure.

6. Placenta

The placenta during pregnancy secretes HCG and progesterone to maintain pregnancy.

7. Thymus

The thymus secretes thymosin for T-lymphocyte maturation. Thymosin is the hormone of the thymus gland that promotes the maturation of T-lymphocytes for immunity.

8. Pineal gland

The pineal gland secretes melatonin, which regulates the sleep-wake cycle.

Tissue / gland	Hormone	Function
Gastrointestinal tract	Gastrin, secretin and cholecystokinin	Coordinate digestion by controlling gastric juice, pancreatic juice and bile release
Kidneys	Erythropoietin	Red blood cell formation
Kidneys	Renin	Blood pressure regulation
Heart	Atrial natriuretic hormone	Lowers blood pressure by increasing salt and water excretion
Placenta	HCG and progesterone	Maintain pregnancy
Thymus	Thymosin	T-lymphocyte maturation
Pineal gland	Melatonin	Regulates the sleep-wake cycle

Feedback Mechanism and Hormone Disorders

LESSON 24

1. Positive feedback mechanism

In positive feedback, the result of a process further increases the same process instead of stopping it, so the response is amplified until the task is completed.

2. Positive feedback in childbirth

A clear example is childbirth: contraction of the uterus pushes the baby against the cervix, stretching it, and this stretching signal causes release of more oxytocin, which makes contractions still stronger, until the baby is delivered and the stimulus ends.

3. Positive feedback in blood clotting

Another example is blood clotting, where activated clotting factors activate more and more factors until the clot seals the wound.

4. Endocrine disorders and their glands

Disorder	Gland
Diabetes	Pancreas
Giantism	Pituitary
Goiter	Thyroid gland — enlargement, usually due to iodine deficiency; not the adrenal medulla
Tetany	Parathyroid

Goiter is not correctly matched with the adrenal medulla. The other pairs are correctly matched: diabetes with pancreas, giantism with pituitary and tetany with parathyroid.

5. Thyroid disorders

Hypothyroidism is under-secretion of thyroxine, causing low metabolic rate, weight gain and sluggishness (cretinism in children, myxedema in adults), while hyperthyroidism is over-secretion, causing high metabolic rate, weight loss, rapid heartbeat and nervousness.

- **Goiter:** enlargement of the thyroid gland, usually caused by iodine deficiency, seen as a swelling in the neck.
- **Cretinism:** hypothyroidism in children.
- **Myxedema:** hypothyroidism in adults.

Animal Behaviour

CHAPTER 5

Stimuli, Reflexes and Innate Behaviour

LESSON 25

1. Stimuli and responses

A stimulus is any detectable change, physical or chemical, in the internal or external environment of an organism, such as light, heat, sound, touch or a chemical. Organisms detect stimuli through receptors and respond through nerve impulses, hormones and muscles or glands. The response may be movement toward or away from the stimulus, a reflex, or a change in internal activity that helps survival.

2. Reflexes

A reflex is a rapid, automatic and involuntary response to a stimulus that does not require conscious thought and follows a fixed reflex arc of receptor, sensory neuron, central nervous system, motor neuron and effector.

- **Knee-jerk reflex:** an example of a reflex in vertebrates.
- **Blinking:** blinking of the eye when an object approaches is an example of a reflex in vertebrates.
- **Withdrawal of the hand:** withdrawal of the hand from a hot object is an example of a reflex in vertebrates.
- **Salivation:** salivation at the sight or smell of food is another common example.

3. Heredity and behaviour

Heredity provides the basic pattern of behaviour through genes, so innate behaviours such as reflexes, instincts and fixed action patterns are inherited and appear without learning. Genes also set the limits within which learning can modify behaviour. Thus behaviour is a product of both inherited tendencies and experience, and closely related animals show similar behaviour because they share similar genes.

4. Innate behaviour and learning behaviour

Innate behaviour is inborn and inherited, is present at birth, needs no experience or practice, and is similar in all members of a species, for example reflexes and nest building. Learning behaviour is acquired during life through experience and practice, differs from individual to individual, and can be changed or improved. Innate behaviour is controlled mainly by genes, while learned behaviour depends on the nervous system and memory.

Feature	Innate behaviour	Learning behaviour
Origin	Inborn and inherited	Acquired during life through experience and practice
Present at birth	Present at birth, needs no experience or practice	Not present at birth
Variation	Similar in all members of a species	Differs from individual to individual, and can be changed or improved
Control	Controlled mainly by genes	Depends on the nervous system and memory
Examples	Reflexes and nest building	Behaviour changed or improved by experience and practice

5. Innate behaviour in terms of reflexes in vertebrates and invertebrates

Innate behaviour is the inherited behaviour performed by an organism from birth in response to a stimulus without prior experience or learning. Its simplest form is the reflex, a rapid automatic response through a reflex arc, such as the knee-jerk, blinking and hand-withdrawal reflexes in vertebrates and the withdrawal of a snail or earthworm on touch in invertebrates. In invertebrates, innate behaviour also appears as kinesis and taxis, for example woodlice moving faster in dry conditions until they reach moisture. In vertebrates, more complex innate patterns develop into instincts and fixed action patterns, but all are built on inherited nervous pathways and need no practice.

6. Instinctive behaviour

Migration of salmon, dances of bees and nest building by birds are classic inborn instinctive patterns performed without learning.

- **Migration of Salmon Fish:** salmon migrate from the sea back to the river of their birth to breed without ever being taught the route. This inborn, inherited pattern is an example of instinctive behaviour.
- **Dances of bees:** a classic inborn instinctive pattern performed without learning.
- **Nest building by birds:** a classic inborn instinctive pattern performed without learning.

7. Construction of an intricate web by a spider as instinct behaviour

Web building by a spider is a classic example of instinct, because a young spider raised in isolation builds a perfect web of its species without ever seeing one or being taught. The spider first lays a bridge thread across a gap, then builds a frame and radial threads from a central hub. It then lays a temporary spiral and replaces it with a sticky capture spiral, testing tension and spacing as it works. The whole sequence is a fixed, inherited pattern released by internal readiness and simple stimuli, and it serves the adaptive purpose of catching prey.

Learned Behaviour — Habituation, Imprinting, Insight and Operant Learning

LESSON 26

1. Habituation — the simplest form of learning

Habituation is the decrease in response to repeated or continuous stimulation. When a stimulus is repeated again and again without any reward or harm, the animal gradually stops responding to it. This simplest form of learning is called habituation.

Habituation is the simplest form of learning in which an animal learns to ignore a repeated stimulus that brings neither benefit nor harm. It saves the animal from wasting energy on useless responses.

- **Repeated stimulus:** a repeated, irrelevant stimulus that brings neither benefit nor harm is gradually ignored.
- **Effect:** the animal learns not to respond or to ignore the stimulus, saving energy that would be wasted on useless responses.

2. Imprinting

Imprinting is a rapid type of learning that occurs during a short sensitive period early in life, in which a young animal fixes its attention on the first moving object it sees and thereafter follows it. Konrad Lorenz showed that newly hatched ducks and goslings follow the first moving object, normally their mother, but they can even be imprinted on a human or an object if that is what they first encounter.

3. How imprinting is adaptive

Imprinting is adaptive because the young stay close to the mother, who protects, warms and guides them to food, greatly increasing their chances of survival.

4. Insight learning

In the 1920s Wolfgang Kohler experimented on chimpanzees and showed insight learning. The chimpanzees suddenly solved problems, for example by joining sticks or piling boxes to reach bananas, without gradual trial and error.

5. Operant learning

B. F. Skinner used the Skinner box to study operant learning in rats and pigeons. The animal learned to press a lever to obtain food as a reward, so its behaviour was shaped by its consequences. The Skinner box was used for operant learning.

Type of learning	Key fact
Habituation	Simplest form of learning — the animal learns to ignore a repeated stimulus that brings neither benefit nor harm
Imprinting	Rapid learning during a short sensitive period early in life — the young animal follows the first moving object it sees
Insight learning	Chimpanzees suddenly solved problems, for example by joining sticks or piling boxes to reach bananas, without gradual trial and error
Operant learning	Studied with the Skinner box in rats and pigeons — behaviour is shaped by its consequences, for example pressing a lever to obtain food as a reward

Taxes, Kinesis and Biological Rhythms

LESSON 27

1. Taxis

Taxis is a directional movement of a whole organism toward or away from a particular stimulus such as light, heat or a chemical. Taxis is a directed movement in which the whole organism moves toward a stimulus (positive taxis) or away from it (negative taxis). In kinesis, by contrast, the response is not directed toward or away from the stimulus.

- **Positive taxis:** movement toward the stimulus, for example moths flying toward light (positive phototaxis) and a sperm moving toward chemicals from the egg (positive chemotaxis).
- **Negative taxis:** movement away from the stimulus, for example a cockroach running away from light and an earthworm moving away from strong light or dryness.

Type	Direction	Examples
Positive taxis	Toward the stimulus	Moths flying toward light (positive phototaxis); a sperm moving toward chemicals from the egg (positive chemotaxis)
Negative taxis	Away from the stimulus	A cockroach running away from light; an earthworm moving away from strong light or dryness

2. Kinesis and other simple behaviour patterns

In kinesis, the response is not directed toward or away from the stimulus. In invertebrates, innate behaviour appears as kinesis and taxis, for example woodlice moving faster in dry conditions until they reach moisture.

Pattern	Nature of response
Taxis	Directed movement — the whole organism moves toward a stimulus or away from it
Kinesis	Response is not directed toward or away from the stimulus
Reflex	Rapid, automatic and involuntary response to a stimulus through a fixed reflex arc
Fixed Action Pattern	A complex innate pattern built on inherited nervous pathways, needing no practice

3. Biological rhythms and the biological clock

The cyclic physiological patterns of activities in an organism that are in response to periodic environmental changes are biological rhythms. Regular cycles of activity controlled by an internal biological clock in response to periodic changes such as day and night are called biological rhythms, and the daily ones are circadian rhythms.

Biological rhythms are the cyclic physiological patterns of activity in an organism that occur in response to periodic environmental changes and are controlled by an internal biological clock.

4. Importance of biological rhythms to man

In man biological rhythms regulate the sleep-wake cycle, body temperature, hormone secretion and alertness in a roughly 24-hour circadian rhythm. Disturbance of these rhythms, for example by night shifts or jet lag, causes tiredness and poor performance, so they are important for health and normal daily life.

- **Sleep-wake cycle:** regulated in a roughly 24-hour circadian rhythm.
- **Body temperature:** regulated in a roughly 24-hour circadian rhythm.
- **Hormone secretion:** regulated in a roughly 24-hour circadian rhythm.
- **Alertness:** regulated in a roughly 24-hour circadian rhythm.
- **Disturbance:** night shifts or jet lag cause tiredness and poor performance.

Social Behaviour — Agonistic Behaviour, Territories and Dominance Hierarchies

LESSON 28

1. Social behaviour

Agonistic behaviour is a type of social behaviour. Agonistic behaviour includes aggression, threat displays, submission and fighting between members of the same species. Because it occurs during social interactions over food, mates or territory, it is a type of social behaviour.

Territorial behaviour in Gorillas is classified as agonistic social behaviour shown in interaction with other members of the group.

2. Agonistic behaviour

Agonistic behaviour is the aggressive and submissive behaviour shown when animals of the same species compete for food, mates, shelter or territory; it includes threat displays, posturing, vocalizations, chasing and sometimes fighting.

- **Threat displays:** part of agonistic behaviour.
- **Posturing:** part of agonistic behaviour.
- **Vocalizations:** part of agonistic behaviour.
- **Chasing:** part of agonistic behaviour.
- **Fighting:** sometimes part of agonistic behaviour.
- **Submission:** agonistic behaviour includes aggression, threat displays, submission and fighting between members of the same species.

3. Territories and the maintenance of social order

By defending a territory, an animal secures enough food and a breeding place, and rivals are kept out mostly by displays rather than injurious fights. Within a group, contests establish a dominance hierarchy or pecking order in which higher-ranking animals get first access to resources and lower-ranking ones submit. In this way agonistic behaviour maintains social order, reduces repeated fighting and stabilizes group life.

Feature	Role in social order
Territory defence	Secures enough food and a breeding place; rivals are kept out mostly by displays rather than injurious fights
Dominance hierarchy / pecking order	Higher-ranking animals get first access to resources and lower-ranking ones submit
Overall effect	Maintains social order, reduces repeated fighting and stabilizes group life

4. How old world monkeys defend their territory

Old world monkeys live in troops and defend their home territory by group displays and aggression against intruding troops. Dominant males give loud calls, show threatening gestures and expose their large canine teeth, and may chase or fight intruders. Such agonistic displays usually settle the dispute without serious injury and keep other troops away from their food and resting sites.

Social Organisation and Communication in Animals

LESSON 29

1. Social organisation in honey bees

In a honey bee colony the workers are sterile females. The workers are females that cannot reproduce. They perform all the colony work such as collecting food, cleaning, feeding larvae and defending the hive, while the queen is the fertile female.

- **Workers:** sterile females that cannot reproduce; they perform all the colony work such as collecting food, cleaning, feeding larvae and defending the hive.
- **Queen:** the fertile female in a honey bee colony.

Member	Fertility	Role
Workers (honey bee)	Sterile females	Collecting food, cleaning, feeding larvae and defending the hive
Queen	Fertile female	The fertile female of the colony

2. How bees communicate about food sources

Honey bees communicate the distance and direction of a food source by performing dances on the comb, as described by Karl von Frisch. A round dance shows that food is near the hive, while a waggle dance, performed in a figure-of-eight with a waggling run, shows a distant source. The angle of the waggle run to the vertical indicates direction with respect to the sun, and the duration of waggling indicates distance.

- **Round dance:** shows that food is near the hive.
- **Waggle dance:** performed in a figure-of-eight with a waggling run; shows a distant source.
- **Direction:** the angle of the waggle run to the vertical indicates direction with respect to the sun.
- **Distance:** the duration of waggling indicates distance.

Dance	What it shows
Round dance	Food is near the hive
Waggle dance	A distant source — performed in a figure-of-eight with a waggling run

3. Communication and group life

Communication about food sources by dances on the comb allows honey bees to share the distance and direction of a food source with other members of the colony. This communication, together with the division of work between the sterile female workers and the fertile queen, is part of the social organisation of the honey bee colony.

Reproduction

CHAPTER 6

Male Reproductive System — Structure and Functions

LESSON 30

1. Male reproductive system

The male reproductive system consists of the testes, epididymis, vas deferens, accessory glands and penis.

- **Testes:** held in the scrotum; they contain seminiferous tubules where sperm are produced and cells that secrete testosterone.
- **Epididymis:** sperm mature and are stored in the coiled epididymis.
- **Vas deferens:** sperm pass through the vas deferens to the urethra.
- **Accessory glands:** the seminal vesicles, prostate gland and Cowper glands add fluids that nourish and protect sperm and form semen.
- **Penis:** the copulatory organ that delivers semen into the female reproductive tract during intercourse.

2. Scrotum and temperature

The scrotum is the sac of skin that holds and protects the testes outside the body cavity. It keeps them about 2-3 degrees Celsius below normal body temperature, which is necessary for normal sperm production.

3. Why the testes descend before birth

The testes develop inside the abdomen of the foetus but descend into the scrotum before birth because sperm production requires a temperature about 2-3 degrees Celsius lower than the internal body temperature. Inside the abdomen the heat would damage the sperm-forming cells and cause sterility. Descent into the cooler scrotum therefore ensures normal spermatogenesis after puberty.

4. Urethra as the urinogenital duct

In the male, the urethra is a common passage for both urine and semen. Urine from the bladder and semen from the vas deferens both leave the body through it, though never at the same time. Because it serves both the urinary and the genital systems, it is called the urinogenital duct.

Male Reproductive Hormones and Spermatogenesis

LESSON 31

1. Principal reproductive hormones of the male

The principal male reproductive hormones are gonadotropin-releasing hormone from the hypothalamus, LH and FSH from the pituitary, and testosterone from the testes.

- **LH:** stimulates the interstitial (Leydig) cells of the testes to secrete testosterone.
- **Testosterone:** responsible for sperm production and the development and maintenance of male secondary sexual characters such as beard, deep voice and body hair.
- **FSH:** acts on the seminiferous tubules and Sertoli cells to promote spermatogenesis.
- **Negative feedback:** through negative feedback, testosterone keeps the pituitary hormones at a steady level, maintaining normal functioning of the system.

2. Role of FSH in sperm production

In males, follicle stimulating hormone (FSH) from the pituitary acts on the seminiferous tubules and Sertoli cells of the testes. It promotes spermatogenesis, that is, the development and maturation of sperm cells.

3. Spermatogenesis

Spermatogenesis is the formation of sperm in the testes. It occurs continuously from puberty and produces four functional sperm from each primary spermatocyte.

Female Reproductive System — Structure and Functions

LESSON 32

1. Female reproductive system

The female reproductive system consists of the ovaries, fallopian tubes, uterus, cervix and vagina.

- **Ovaries:** produce eggs and the hormones estrogen and progesterone, and release one egg each month at ovulation.
- **Fallopian tubes:** the fallopian tubes (oviducts) carry the egg toward the uterus and are the usual site of fertilization.
- **Uterus:** a muscular organ whose lining, the endometrium, receives the implanted embryo and nourishes the developing foetus during pregnancy.
- **Cervix:** the narrow lower end of the uterus opening into the vagina.
- **Vagina:** receives sperm during intercourse and forms the birth canal during childbirth.

2. Ovarian cycle and the mature follicle

During the ovarian cycle a primary follicle develops into a secondary follicle and then into the mature follicle. This mature follicle is called both the Graafian follicle and the vesicular follicle, so both names are correct. The ovarian cycle includes changes in ovarian follicles to secondary follicle and finally to Graafian follicle.

3. Testes and ovaries

Feature	Testes	Ovaries
Position	Male gonads, located in the scrotum	Female gonads, located in the pelvic cavity
Products	Produce sperm and testosterone	Produce eggs (ova) and estrogen and progesterone

Menstrual Cycle and Its Hormonal Regulation

LESSON 33

1. Phases of the menstrual cycle

The menstrual cycle of about 28 days has three main phases.

- **Menstrual phase (days 1–5):** the thickened endometrium breaks down and is shed with blood through the vagina because no pregnancy has occurred.
- **Proliferative or follicular phase (days 6–14):** FSH develops an ovarian follicle, and estrogen from the follicle repairs and thickens the endometrium, ending with ovulation around day 14.
- **Secretory or luteal phase (days 15–28):** the corpus luteum secretes progesterone, which makes the endometrium thick, vascular and secretory for possible implantation; if fertilization fails, hormone levels fall and the cycle repeats.

2. Why the inner uterine wall thickens in the secretory phase

During the secretory phase, progesterone from the corpus luteum makes the inner uterine wall (endometrium) thicken, become rich in blood vessels and glands that secrete nutrients. This prepares a soft, nourishing bed in which a fertilized egg can implant and grow. If fertilization does not occur, progesterone falls and this thickened lining is shed during menstruation.

3. Hormones that regulate the menstrual cycle

The menstrual cycle is regulated by FSH and LH from the pituitary together with estrogen and progesterone from the ovary. Androgens are male hormones and do not primarily regulate the female menstrual cycle.

- **Estrogen:** under the influence of FSH the growing Graafian follicle secretes estrogen. Estrogen causes growth and repair of the uterine lining during the first half of the menstrual cycle.
- **Progesterone:** estrogen and progesterone are the two key ovarian hormones that regulate the menstrual cycle, controlling follicle growth, ovulation and preparation of the uterus.
- **Testosterone:** the main male hormone and is not a key regulator of the cycle.

Hormone	Role in the cycle
FSH	Develops an ovarian follicle; induces the Graafian follicle to produce estrogen
LH	Regulates the cycle together with FSH
Estrogen	Growth and repair of the uterine lining during the first half of the cycle
Progesterone	Makes the endometrium thick, vascular and secretory in the secretory phase

4. Female hormonal system

The female hormonal system is considered better, or more finely regulated, because it works in a precise cyclic manner with feedback control between the pituitary and the ovaries. Estrogen and progesterone rise and fall in a coordinated sequence that controls follicle growth, ovulation, preparation of the uterus and, if pregnancy occurs, its maintenance. This regular hormonal cycle prepares the body every month for a possible pregnancy, a level of cyclic coordination not found in the male.

Gametogenesis, Infertility and Reproductive Health

LESSON 34

1. Spermatogenesis and oogenesis

Feature	Spermatogenesis	Oogenesis
Meaning	The formation of sperm in the testes	The formation of eggs in the ovaries
Timing	Occurs continuously from puberty	Begins before birth and occurs cyclically
Products	Produces four functional sperm from each primary spermatocyte	Produces only one functional ovum and polar bodies from each primary oocyte

2. Causes of infertility

- **Male infertility:** low sperm count, poor sperm motility or abnormal sperm, blocked vas deferens, hormonal imbalance, undescended testes and infections.
- **Female infertility:** ovulation disorders such as PCOS, blocked fallopian tubes, endometriosis, uterine abnormalities and hormonal imbalance.
- **Both sexes:** age, smoking and general ill health can contribute to infertility in both sexes.

Polycystic ovary syndrome (PCOS) disturbs ovulation, endometriosis damages the reproductive organs, and blocked fallopian tubes prevent the egg and sperm from meeting. Any of these can cause female infertility.

3. In-vitro fertilization (IVF)

In IVF, eggs are fertilized by sperm outside the body in the laboratory and the resulting embryo is placed in the uterus. It is an assisted reproductive technique used to help couples who cannot conceive naturally, and it is a method used to treat infertility.

4. Miscarriage

A miscarriage is the natural, spontaneous loss of a pregnancy before the foetus can survive outside the uterus. In medical terms it is called spontaneous abortion, whereas induced abortion is a deliberate termination.

5. Sexually transmitted diseases

- **Syphilis:** a sexually transmitted disease caused by the bacterium Treponema pallidum, transmitted through sexual contact. It produces sores and rashes and, if untreated, can damage the heart and nervous system, but it can be cured with antibiotics.
- **Gonorrhea:** a bacterial sexually transmitted disease caused by Neisseria gonorrhoeae. Its symptoms include a burning sensation and pain during urination and a thick yellow-green discharge from the urethra or vagina. If untreated it can inflame the reproductive organs and cause infertility, and some infected persons, especially women, may show few early symptoms.

Development and Aging

CHAPTER 7

Fertilization, Cleavage and the Blastocyst

LESSON 35

1. Fertilization and the zygote

Fertilization normally takes place in the upper part of the fallopian tube (oviduct), where the sperm meets the egg released at ovulation. The fertilized egg then travels to the uterus for implantation.

At fertilization the haploid nucleus of the sperm fuses with the haploid nucleus of the egg. The resulting zygote therefore contains a complete diploid set of DNA, half from the mother and half from the father, so the genetic material of a fertilized egg is a combination of both the mother's and father's DNA.

2. Stages of embryonic development

After fertilization, the stages of embryonic development are cleavage, implantation, gastrulation, and organogenesis. The zygote undergoes cleavage to form a blastocyst, which implants in the uterus. Gastrulation then forms the three germ layers, and organogenesis forms the organs from these layers.

3. Cleavage and the morula

After fertilization the zygote undergoes repeated mitotic divisions called cleavage, first forming a solid morula and then the blastula, which in mammals is called the blastocyst.

- **Morula:** the early embryonic stage formed by cleavage of the zygote, consisting of a solid ball of small cells called blastomeres. Its surface looks like a mulberry fruit (Morula is the Latin name related to mulberry), so it is named morula. It later develops a cavity and changes into the blastula or blastocyst.

4. Meroblastic cleavage in birds and reptiles

In birds and reptiles the egg contains a very large amount of yolk, and cleavage furrows cannot pass through the dense yolk. Only the small disc of cytoplasm on top of the yolk, the blastodisc, divides. Because cleavage is partial and incomplete, involving only part of the egg, it is called meroblastic cleavage.

5. Blastula and blastocyst — segmentation cavity

Cleavage creates a fluid-filled space inside, the segmentation cavity or blastocoel, formed as the blastomeres separate and fluid accumulates. In the mammalian blastocyst the cells are arranged as an outer layer, the trophoblast, which will form the placenta and membranes, and an inner cell mass attached at one side, which will form the embryo.

- **Trophoblast:** the outer layer of the blastocyst; it mainly contributes to the placenta and foetal membranes.
- **Inner cell mass:** gives rise to the embryo proper.
- **Segmentation cavity:** the fluid-filled space inside, also called the blastocoel; it allows cell movements and rearrangement during the next stage, gastrulation, and in mammals gives the blastocyst the form that implants in the uterine wall.

Implantation, Gastrulation and Embryonic Induction

LESSON 36

1. Implantation and development in the uterus

Human embryonic development occurs in the uterus. After fertilization in the fallopian tube, the embryo implants in the wall of the uterus. It then develops there, nourished through the placenta, until birth.

The fertilized egg divides during its journey down the fallopian tube and forms a blastocyst. About six to seven days, roughly one week, after fertilization the blastocyst implants in the endometrium of the uterus.

2. Blastula and gastrula

Feature	Blastula	Gastrula
Structure	An early stage consisting of a hollow ball of cells (blastoderm) around a fluid-filled cavity, the blastocoel or segmentation cavity	The later stage formed by gastrulation, in which cells migrate inward
Germ layers and gut	Has no germ layers or gut	Has three germ layers (ectoderm, mesoderm and endoderm) and a primitive gut, the archenteron
Significance	Early stage after cleavage	Marks the beginning of tissue and organ formation

3. Embryonic induction — how tissues influence other tissues

Embryonic tissues influence neighbouring tissues by the process of induction, in which one tissue releases chemical signals that direct adjacent cells to differentiate in a particular way. For example, the chordamesoderm induces the overlying ectoderm to form the neural plate and neural tube. Such interactions, controlled by signal molecules and gene switching, coordinate the orderly development of organs.

4. Organizers and primary and secondary induction

An organizer is a group of embryonic cells that influences the development of surrounding tissues and organizes the body plan, such as the dorsal lip of the blastopore in amphibians described by Spemann.

- **Primary induction:** the first inductive interaction, in which the chordamesoderm induces the ectoderm to form the nervous system during gastrulation.
- **Secondary induction:** occurs later between tissues formed after gastrulation, for example the optic vesicle inducing the overlying ectoderm to form the lens of the eye.

Germ Layers, Differentiation and Organogenesis

LESSON 37

1. Cell differentiation

The process of cell differentiation involves the specialization of cells for different functions. In differentiation, cells that have the same genes switch on different genes and become specialized in structure and function, for example into muscle, nerve or blood cells. This specialization builds the different tissues and organs of the body.

2. Organogenesis and the neural tube

During organogenesis the ectoderm folds to form the neural tube, which develops into the brain and spinal cord. This process, called neurulation, is one of the earliest events in organ formation. The formation of the neural tube occurs during organogenesis.

3. Tissues and organs formed from the three germ layers

- **Ectoderm:** forms the epidermis of the skin with hair, nails and skin glands, the whole nervous system including brain and spinal cord, the sensory epithelia of the eye, ear and nose, the lens and retina, and the enamel of teeth.
- **Mesoderm:** forms muscles, bones and cartilage, connective tissues, the heart and blood vessels, blood, kidneys and ureters, the reproductive organs, the dermis of skin and the lining of the body cavity.
- **Endoderm:** forms the lining of the digestive tract and respiratory tract, and the glands derived from them including the liver, pancreas, thyroid and thymus, as well as the lining of the urinary bladder.

Germ layer	Tissues and organs formed
Ectoderm	Epidermis of the skin with hair, nails and skin glands; whole nervous system including brain and spinal cord; sensory epithelia of the eye, ear and nose; lens and retina; enamel of teeth
Mesoderm	Muscles, bones and cartilage; connective tissues; heart and blood vessels; blood; kidneys and ureters; reproductive organs; dermis of skin; lining of the body cavity
Endoderm	Lining of the digestive tract and respiratory tract; glands derived from them including liver, pancreas, thyroid and thymus; lining of the urinary bladder

4. Role of the nucleus and cytoplasm in controlling development

Experiments show that both the nucleus and the cytoplasm control development. In nuclear transplantation experiments, for example Briggs and King transplanted a nucleus from an early frog embryo into an enucleated egg and a normal tadpole developed, proving the nucleus carries the complete genetic instructions for development. Gurdon later obtained adult frogs from nuclei of tadpole intestinal cells, showing that differentiated cells retain all genes.

On the other hand, the egg cytoplasm contains maternal determinants distributed unevenly, and experiments such as constricting a fertilized egg show that regions lacking certain cytoplasmic substances fail to form particular structures. Development is therefore controlled by the genes of the nucleus acting in response to signals and substances in the surrounding cytoplasm.

Foetal Development, the Placenta and the Trimesters

LESSON 38

1. The placenta

The placenta transfers oxygen and nutrients from the mother to the embryo and removes carbon dioxide and other waste products from the foetal blood. It also secretes hormones, such as progesterone, that maintain pregnancy.

- **Providing oxygen and nutrients:** the placenta is responsible for providing oxygen and nutrients to the developing embryo.
- **Removing waste products:** the placenta is responsible for removing waste products from the developing embryo.
- **Producing hormones:** the placenta is responsible for producing hormones that maintain the pregnancy.

2. Development in the first, second and third trimesters

- **First trimester (weeks 1-12):** fertilization, cleavage, implantation and gastrulation occur, the three germ layers form, and organogenesis builds all major organs; the heart starts beating, limb buds appear and by the end the embryo is called a foetus.
- **Second trimester (weeks 13-26):** the foetus grows rapidly, bones ossify, hair and facial features form, movements are felt by the mother and the sex can be identified, while the organs mature.
- **Third trimester (weeks 27-40):** growth and weight gain are greatest, the lungs mature with surfactant production, the foetus turns head-down, and it becomes capable of independent life at birth.

3. Monozygotic twins

Monozygotic twins develop from a single fertilized egg that splits into two embryos early in development. Because both embryos come from the same zygote, they carry identical chromosomes, including the same sex chromosomes (both XX or both XY). They are therefore always of the same sex and are called identical twins.

4. Maternal derived abnormalities

Maternal factors can disturb embryonic development, showing that a healthy maternal environment is essential for normal development.

- **Rubella:** Rubella (German measles) virus crossing the placenta in early pregnancy can cause deafness, cataracts and heart defects in the baby.
- **Abnormal neural tube:** failure of the neural tube to close properly produces abnormalities such as spina bifida and anencephaly, commonly linked with deficiency of folic acid in the mother.
- **Thyroid gland:** maternal thyroid disorders, especially iodine deficiency and hypothyroidism, impair the baby's brain and skeletal development and can cause cretinism.
- **Limb development:** drugs and harmful substances taken by the mother, such as thalidomide, can disturb limb development and produce short or missing limbs.

Aging — Accumulation of Mutations and Decline in Function

LESSON 39

1. Aging

Aging is characterized by the accumulation of genetic mutations. With advancing age, DNA damage and mutations accumulate in cells while repair becomes less efficient.

2. Causes and effects of aging

- **Accumulation of genetic mutations:** DNA damage and mutations accumulate in cells with advancing age.
- **Breakdown of cellular processes:** the accumulation contributes to the gradual breakdown of cellular processes.

- **Decline in tissue and organ function:** the breakdown leads to the decline in tissue and organ function seen in aging, as repair becomes less efficient.

This gradual decline is the pattern seen in aging: mutations accumulate, repair becomes less efficient, cellular processes break down gradually, and tissue and organ function declines.

Chromosome and DNA

CHAPTER 8

Chromosomes — Structure and the Chromosomal Theory of Inheritance

LESSON 40

1. Chromosomal theory of inheritance

The chromosomal theory of inheritance was proposed by Sutton and Boveri in 1902. It states that genes are located on chromosomes and that chromosomes are the carriers of heredity from one generation to the next. The nucleus is the storehouse of this hereditary material.

Walter Sutton and Theodor Boveri observed that chromosomes are distributed during meiosis and fertilization in the same pattern as Mendel's hereditary factors. This parallel behaviour led them to propose the theory.

2. Parallel behaviour of chromosomes and Mendel's factors

- **Occur in pairs:** chromosomes and Mendel's factors both occur in pairs.
- **Segregation:** both segregate during meiosis so that each gamete receives one member of each pair.
- **Restoration at fertilization:** both are restored at fertilization when the sperm and egg nuclei fuse.
- **Equal contribution:** both parents make an equal hereditary contribution through their gametes, each gamete carrying one copy of each homologous chromosome.

The behaviour of chromosomes in meiosis thus explains Mendel's laws of segregation and independent assortment. Each parent makes an equal chromosomal contribution to the offspring.

Feature	Chromosomes	Mendel's factors
Occurrence	Occur in pairs	Occur in pairs
During meiosis	Segregate, one member to each gamete	Segregate, one member to each gamete
At fertilization	Pairs restored when sperm and egg nuclei fuse	Pairs restored in the offspring

3. Chemical structure of a chromosome

A chromosome is made of chromatin, which chemically consists of DNA, histone and non-histone proteins, and a small amount of RNA.

- **DNA:** about 40 percent of the chromosome; the hereditary material carrying the genes.
- **Protein:** about 60 percent, mainly basic histones around which the DNA is wound to form nucleosomes.
- **RNA:** present in a small amount.

At division the chromatin condenses into the short, thick chromosome. This packing allows the very long DNA molecule to fit inside the nucleus and to be distributed accurately to daughter cells.

4. Sister chromatids and the centromere

After replication, each chromosome consists of two identical sister chromatids joined at the centromere. These chromatids separate during cell division so that each daughter cell receives one copy. At division the condensed chromosome therefore shows two sister chromatids joined at a centromere, with telomeres at the ends.

5. Nucleosome and chromatin packing

A nucleosome is the basic unit of chromatin packing, formed by about two loops of duplex DNA wrapped around a core of eight histone proteins, a histone octamer. A chain of nucleosomes gives chromatin its beads-on-a-string appearance.

- **Histone octamer:** a core of eight histone proteins.
- **DNA wrapping:** about two loops of duplex DNA wrapped around the octamer.
- **Nucleosome:** the complex of histone octamer and two loops of duplex DNA.

DNA Structure, Genes and Gene Expression

LESSON 41

1. Gene, genome and locus

A gene is a specific length of DNA that carries the coded information for making one particular polypeptide chain. A small part of DNA which has information to synthesize a specific polypeptide chain is a gene.

- **Gene:** a specific length of DNA carrying the coded information for one particular polypeptide chain.
- **Genome:** the total DNA of the cell.
- **Locus:** the position of a gene on a chromosome.

2. Why DNA is a negatively charged molecule

DNA is negatively charged because each nucleotide of its backbone contains a phosphate group, and phosphate groups carry a negative charge at normal cellular pH. With thousands of phosphate groups along its length, the whole DNA molecule has a strong overall negative charge. This charge helps DNA associate with positively charged histone proteins to form chromatin.

- **Phosphate groups:** each nucleotide of the backbone contains a phosphate group carrying a negative charge at normal cellular pH.
- **Overall charge:** thousands of phosphate groups give the whole molecule a strong overall negative charge.
- **Association with histones:** the negative charge helps DNA associate with positively charged histone proteins to form chromatin.

3. Gene expression

Gene expression is the overall process by which the information stored in a gene is used to produce a functional product, usually a protein. It includes transcription of the gene into mRNA and translation of mRNA into a polypeptide.

4. Genes in prokaryotic and eukaryotic cells

Prokaryotic and eukaryotic genes differ in structure.

Feature	Prokaryotic genes	Eukaryotic genes
Structure	Continuous, without introns	Split into coding exons and non-coding introns
Organisation	Often grouped in operons	Each gene is usually transcribed separately
DNA form and location	Single circular DNA molecule in the nucleoid	Linear DNA packed with histones in a membrane-bound nucleus
Transcription and translation	Occur together in the cytoplasm	Primary transcript must be processed before translation

Replication of DNA

LESSON 42

1. Models of DNA replication

In the conservative model, the parental double helix was proposed to remain completely intact and an entirely new double helix to be made. Actual DNA replication was later proved by Meselson and Stahl to be semi-conservative, not conservative.

2. Why replication is semi-conservative

Replication is called semi-conservative because each daughter DNA molecule formed consists of one old (parental) strand and one newly synthesized strand. In this way half of the original molecule is conserved in each new molecule. The Meselson and Stahl experiment with nitrogen isotopes proved this mode of replication.

Meselson and Stahl proved this mechanism by growing bacteria on heavy nitrogen (N-15) and then on light nitrogen (N-14) and finding DNA of intermediate density after one generation.

3. The semi-conservative process

In semi-conservative replication the two strands of the parental double helix separate, and each acts as a template for a new complementary strand, so each daughter molecule keeps one old strand and gains one new strand.

- **Unwinding:** helicase unwinds the helix at the origin and a replication fork forms, with SSB proteins holding the strands apart.
- **Priming:** primase lays a short RNA primer.
- **Strand building:** DNA polymerase III adds complementary nucleotides continuously on the leading strand and in Okazaki fragments on the lagging strand.
- **Finishing:** polymerase I and ligase remove primers and seal the joins.

4. Leading and lagging strands

Because DNA polymerase can synthesize DNA only in the 5' to 3' direction and the two template strands are antiparallel, the two new strands are made differently.

- **Leading strand:** synthesized continuously in the 5' to 3' direction, in the same direction as the movement of the replication fork, from a single primer.
- **Lagging strand:** synthesized discontinuously, away from the fork, as short Okazaki fragments that are later joined by DNA ligase.

Feature	Leading strand	Lagging strand
Direction of synthesis	5' to 3', same direction as fork movement	5' to 3', away from the fork
Continuity	Continuous	Discontinuous, in short Okazaki fragments
Primer	Single primer	Primer for each fragment
Joining	No fragments to join	Fragments later joined by DNA ligase

5. Single-strand binding proteins

Single-strand binding proteins (SSB) attach to the separated single strands of DNA at the replication fork. They keep the strands apart, prevent them from re-joining and protect them from damage until new strands are synthesized. A protein which prevents the re-binding of complementary strands during replication at the fork is SSB.

6. Enzymes involved in replication

- **Helicase:** unwinds and separates the two strands.
- **Primase:** lays down a short RNA primer.
- **DNA polymerase III:** builds the new DNA strand.
- **DNA polymerase I:** removes the primer and fills the gap.
- **Ligase:** joins the Okazaki fragments.
- **SSB proteins:** keep the single strands separated.
- **Topoisomerase:** relieves the twisting strain ahead of the fork.

Transcription and mRNA Processing

LESSON 43

1. Transcription

Transcription is the first step of gene expression, in which the information of one gene on DNA is copied into a complementary messenger RNA by the enzyme RNA polymerase.

- **Binding:** the polymerase binds at the promoter and unwinds a short length of DNA.
- **Reading the template:** the polymerase reads the template strand in the 3' to 5' direction.
- **Building mRNA:** complementary RNA nucleotides are joined, with uracil in place of thymine, to build mRNA in the 5' to 3' direction.
- **Termination:** transcription continues until a terminator sequence is reached, when the mRNA is released.

In eukaryotes the transcript is then capped, tailed and spliced before it leaves the nucleus for translation.

2. Introns are transcribed but not translated

Introns are transcribed, because RNA polymerase copies the whole gene, including both exons and introns, into the primary mRNA transcript. However, introns are not involved in translation, because they are cut out during splicing before the mature mRNA leaves the nucleus. Only the joined exons are read by the ribosome to make the protein.

- **Transcription:** the whole gene, exons and introns, is copied into the primary transcript.
- **Splicing:** introns are cut out before the mature mRNA leaves the nucleus.
- **Translation:** only the joined exons are read by the ribosome.

3. Processing of mRNA before it leaves the nucleus

Before leaving the nucleus, eukaryotic mRNA is processed in three ways. This mature mRNA is stable, is protected from degradation, and can be translated in the cytoplasm. Only after this processing does the mRNA pass through the nuclear pores to the ribosomes.

- **5' cap:** a cap is added at the 5' end.
- **Poly-A tail:** a poly-A tail is added at the 3' end.
- **Splicing:** the introns are spliced out so that the exons join together.

Change	Where / what happens
Cap	Added at the 5' end

Change	Where / what happens
Poly-A tail	Added at the 3' end
Splicing	Introns removed, exons joined together

Translation — tRNA and the Ribosome

LESSON 44

1. Sites at the ribosome during translation

The large subunit of the ribosome has three sites for tRNA during translation. The ribosome moves along the mRNA codon by codon, linking amino acids into a chain at these sites.

- **A site:** the aminoacyl site; receives the incoming tRNA carrying the next amino acid.
- **P site:** the peptidyl site; holds the tRNA carrying the growing polypeptide chain.
- **E site:** the exit site; where the empty tRNA leaves the ribosome.

Site	Full name	Function
A	Aminoacyl site	Receives the incoming tRNA carrying the next amino acid
P	Peptidyl site	Holds the tRNA carrying the growing polypeptide chain
E	Exit site	Empty tRNA leaves the ribosome

2. Structure of tRNA

A tRNA molecule is folded into a clover-leaf shape with several important regions.

- **Amino acid attachment site:** at the 3' end is the CCA end, which carries a specific amino acid.
- **Anticodon loop:** at the opposite end; its three-base anticodon pairs with the complementary codon on mRNA.
- **D loop and T loop:** help the molecule bind correctly to the ribosome and to its specific activating enzyme.

3. Function of tRNA

The tRNA delivers the correct amino acid to the growing chain at the ribosome. Its anticodon ensures pairing with the complementary codon on mRNA, its 3' CCA end carries the specific amino acid, and its D and T loops help it bind correctly to the ribosome and to its specific activating enzyme.

Genetic Code, Mutations and Gene Regulation

LESSON 45

1. The genetic code

The genetic code is the set of rules by which triplet sequences of bases in DNA and mRNA specify the amino acids of a protein. It is the set of rules used to store the genetic information within DNA for particular protein synthesis.

- **Genetic code:** the set of rules by which triplet sequences of bases in DNA and mRNA specify the amino acids of a protein.
- **Codon:** a single triplet within this code.
- **Anticodon:** the matching triplet on tRNA.

2. Mutation is not always harmful

Mutation is not always harmful.

- **Neutral mutations:** many mutations are neutral because they occur in non-coding DNA or do not change the amino acid produced, so they have no visible effect.
- **Harmful mutations:** some mutations are harmful and cause genetic disorders.
- **Beneficial mutations:** a few are beneficial, giving the organism an advantage in survival and reproduction.

Such useful variations are the raw material of evolution and of plant and animal breeding.

Type of effect	Outcome
Neutral	Occurs in non-coding DNA or does not change the amino acid; no visible effect
Harmful	Causes genetic disorders
Beneficial	Gives an advantage in survival and reproduction; raw material of evolution and breeding

3. Gene regulation

Gene regulation is the turning ON and OFF of genes so that a cell makes only the proteins it needs, at the right time and in the right amount.

4. Gene regulation in prokaryotes

In prokaryotes regulation occurs mainly at transcription through operons, where a repressor protein blocks the operator to switch genes off, and removal of the repressor or binding of an activator switches them on, as in the lac operon.

- **Repressor:** a protein that blocks the operator to switch genes off.
- **Switching on:** removal of the repressor or binding of an activator switches genes on.
- **Example:** the lac operon.

5. Gene regulation in eukaryotes

In eukaryotes regulation occurs at many levels: chromatin opening, control of transcription by regulatory proteins, mRNA processing, mRNA breakdown and control of translation. Such regulation allows cells with identical genes to differentiate into different types and to respond to changing conditions.

- **Chromatin opening:** regulation at the level of chromatin.
- **Transcription control:** control of transcription by regulatory proteins.
- **mRNA levels:** mRNA processing and mRNA breakdown.
- **Translation control:** control of translation.

Inheritance

CHAPTER 9

Mendel's Laws — Monohybrid, Dihybrid, Segregation and Independent Assortment

LESSON 46

1. Genetics, genes and alleles

Genetics is the branch of biology that studies heredity and variation. Organisms with the same genotype can look different when they grow under different conditions of food, light, temperature or water. Such variation is environmental and is not inherited.

- **Gene:** a specific part of DNA that controls one hereditary character.
- **Allele:** an alternative form of a gene present at the same locus on homologous chromosomes.
- **Genotype and environment:** the genotype sets the hereditary characters, while different environments can produce phenotypic variation among organisms of the same genotype.

2. Monohybrid and dihybrid crosses

Cross	What it studies
Monohybrid cross	One contrasting trait at a time
Dihybrid cross	Two traits at the same time

3. Law of segregation — purity of gametes

According to the law of segregation, the two alleles of a gene separate during gamete formation, so each gamete receives only one allele and remains pure for that character. A gamete never carries both alleles together, therefore this law is called the law of purity of gametes.

4. Dihybrid cross — inheritance of two traits

Inheritance of two traits is studied by a dihybrid cross, for example a cross between pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy). All F1 plants are heterozygous (RrYy) with round yellow seeds, showing that round and yellow are dominant.

Each gene pair segregates and assorts independently, producing four types of gametes (RY, Ry, rY, ry) from each parent. The appearance of new combinations, round green and wrinkled yellow, proves the law of independent assortment.

F2 phenotype	Ratio
Round yellow	9
Round green	3
Wrinkled yellow	3
Wrinkled green	1

On self-crossing, F1 gives F2 in the ratio 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.

Test Cross and Back Cross

LESSON 47

1. Test cross

A test cross is a cross of an individual of unknown genotype showing the dominant phenotype with a homozygous recessive individual, to find out its genotype.

- **Purpose:** to find out the genotype of an individual showing the dominant phenotype.
- **Partner in the cross:** a homozygous recessive individual.

2. Back cross

A back cross is a cross of the F1 hybrid with either of its parents.

Cross	Definition
Test cross	An individual of unknown genotype showing the dominant phenotype crossed with a homozygous recessive individual, to find out its genotype
Back cross	The F1 hybrid crossed with either of its parents

3. Related terms

- **Mutant:** an organism or gene that has undergone a sudden heritable change called mutation.
- **Karyotype:** the complete set of chromosomes of a cell shown by number, size and shape.

Incomplete Dominance

LESSON 48

1. Incomplete dominance — partial dominance

In incomplete dominance the dominant allele cannot completely suppress the recessive allele in the heterozygote, so the offspring shows a phenotype midway between the two parents. Because dominance is only partial and not complete, it is also called partial dominance.

- **Heterozygote:** shows a blended, intermediate phenotype.
- **Example in Mirabilis:** pink flowers are produced from red and white parents.

2. Incomplete dominance and codominance compared

Incomplete dominance	Codominance
The heterozygote shows a blended, intermediate phenotype	Both alleles are fully and separately expressed in the heterozygote, as in blood group AB

Codominance

LESSON 49

1. Codominance

In codominance both alleles are fully and separately expressed in the heterozygote. This is different from incomplete dominance, in which the heterozygote shows a blended, intermediate phenotype.

Pattern	Expression in the heterozygote
Incomplete dominance	A blended, intermediate phenotype
Codominance	Both alleles fully and separately expressed

2. Codominance in the ABO system

In the ABO system, IA and IB are co-dominant to each other. The genotype IAIB gives group AB, in which both antigens A and B are present, so both alleles are expressed together in the same person.

- **IAIB:** blood group AB — both antigens A and B are present.
- **Co-dominant male:** a male with the co-dominant blood group has group AB (IAIB).

Multiple Alleles and Blood Groups

LESSON 50

1. Multiple alleles

Multiple alleles are three or more alternative forms of the same gene present in a population, such as IA, IB and i for blood groups. Any person carries only two of these alleles.

The ABO blood group system is an example of inheritance explained by multiple alleles on an autosomal chromosome. The ABO gene on an autosome has three alleles, IA, IB and i. Different combinations of these alleles produce blood groups A, B, AB and O.

2. The ABO blood system

The ABO system is controlled by a single gene with three alleles: IA and IB are co-dominant to each other and both are dominant over i.

Blood group	Genotype	Antigen
A	IAIA or IAi	A
B	IBIB or IBi	B
AB	IAIB	Both antigens A and B are present
O	ii	No antigen

- **Isoagglutinogens:** the antigens (A and B) present on the surface of red blood cells that cause agglutination with the opposite antibody.

3. ABO cross — group A x group B parents

A cross between a group A father (IAi) and a group B mother (IBi) can produce children of all four groups: IAIB (AB), IAi (A), IBi (B) and ii (O), each with a 25 percent chance. Two parents neither of whom is AB or O can still have AB and O children.

4. Group A parents with a group O child

A child with blood group O proves that both group A parents are heterozygous, IAi. In the cross IAi x IAi, one out of four children, that is 0.25, receives i from both parents and has blood group O. The probability that the next child will be a boy who has blood group O is 0.25 as printed for this cross.

5. Codominant AB cross through two generations

The co-dominant blood group of the male is AB (IAIB) and the female is homozygous A (IAIA). Their children receive IA from the mother and either IA or IB from the father, so all their children are either IAIA (group A) or IAIB (group AB), each with 50 percent chance; their male child is therefore either group A (IAIA) or group AB (IAIB).

When this son marries a female with the recessive blood group O (ii), she gives i to every child: if the son is IAIA, all children are IAi, that is 100 percent group A; if the son is IAIB, half the children are IAi (group A) and half are IBi (group B). The grandchildren can only be group A or group B, and group O or AB is not possible in this cross.

Son's genotype	Cross with ii	Children
IAIA	IAIA x ii	All IAi — 100 percent group A
IAIB	IAIB x ii	Half IAi (group A) and half IBi (group B)

6. Blood group systems

Apart from the well known ABO and Rh systems, the International Society of Blood Transfusion has recognised more than 30 blood group systems in man. Each system is controlled by its own gene or group of genes and its antigens must be considered during transfusion.

7. Rh incompatibility

An Rh-negative mother carrying an Rh-positive foetus may be sensitised by foetal red cells entering her blood at delivery, and she then makes anti-Rh antibodies. In a later pregnancy these antibodies cross the placenta and destroy the red blood cells of an Rh-positive foetus, causing erythroblastosis foetalis, severe anaemia, jaundice and even death of the foetus. Repeated sensitisation also makes transfusion and later pregnancies risky for the mother.

- **Erythroblastosis foetalis:** the disease of a newborn Rh-positive baby of an Rh-negative mother, in which maternal anti-Rh antibodies destroy the baby's red blood cells.

Sex Linkage and Other Inheritance Patterns

LESSON 51

1. Sex determination in humans

In humans females are XX and males are XY. The mother always passes an X chromosome to the child, while the father passes either X, producing a daughter (XX), or Y, producing a son (XY), so the father's sperm determines the sex of the baby in roughly a 50:50 ratio.

2. Why the human male is heterogametic

A human male has two different sex chromosomes, X and Y, so during meiosis he produces two types of sperms: half carrying X and half carrying Y. Because he produces more than one type of gamete for sex, the male is called heterogametic, while the female (XX) produces only X-carrying eggs and is homogametic.

3. Autosomes and sex chromosomes

Chromosomes	Features
Autosomes	Alike in males and females and do not determine sex
Sex chromosomes	Differ between the sexes and determine sex

4. ZZ and ZW sex determination

In birds, butterflies and some reptiles the sex chromosomes are called Z and W instead of X and Y. The male is homogametic with ZZ chromosomes and produces only Z sperms, while the female is heterogametic with ZW chromosomes and produces Z and W eggs, so in these animals the egg determines the sex of the offspring.

5. Grasshopper — XX and XO

In grasshopper the female has two X chromosomes (XX) while the male has only one X chromosome (XO) and no second sex chromosome. The female number of chromosomes is 24, and the male therefore has one chromosome less, 23 instead of 24.

6. X-linked and Y-linked traits

Trait	Features
X-linked trait	Controlled by a gene on the X chromosome and can appear in both sexes
Y-linked trait	Controlled by a gene on the Y chromosome and appears only in males

- **Holandric genes:** present only on the non-homologous part of the Y chromosome. They pass directly from father to son only, and are always expressed because males are hemizygous for them.

7. X-linked inheritance — colour blindness

Genes present on the X chromosome show X-linked inheritance, for example colour blindness: a carrier mother (XN Xn) crossed with a normal father (XN Y) gives normal daughters, half of them carriers, normal sons, and colour-blind sons (Xn Y) in equal proportions. Sons therefore get X-linked recessive disorders from their carrier mothers, while daughters are affected only when they receive the allele from both parents.

- **Colour-blind son:** a boy gets his only X chromosome from his mother, so his mother must carry the recessive colour-blind allele n on one X chromosome. A carrier mother XN Xn is phenotypically normal but passes Xn to half of her sons.
- **Parents of a colour-blind son and a non-carrier daughter:** the father is normal and the mother is a carrier. The son receives his X chromosome from his mother, so the mother must carry the colour-blind allele. The daughter receives her father's X chromosome, so a normal father gives her a normal X and she is a non-carrier when she also receives a normal X from her mother.
- **Protanopia:** red colour blindness, in which a person cannot perceive red colour.
- **Dichromacy:** a form of colour blindness in which a person can perceive only two of the three primary colours because one type of cone cells is missing or faulty, for example red blindness (protanopia) or green blindness (deuteranopia).
- **Monochromacy:** complete colour blindness, in which the person cannot distinguish colours at all and sees everything in black, white and shades of grey.

8. Haemophilia

Haemophilia is an X-linked recessive disorder, and a male has only one X chromosome, so a single recessive allele on that X is expressed and he becomes haemophilic. A female has two X chromosomes, so she is usually only a carrier when one X carries the allele, and she becomes haemophilic only in the rare case when both X chromosomes carry it.

9. Sex-influenced traits

A sex-influenced trait is an autosomal trait whose expression differs in males and females because of sex hormones, so the same genotype behaves as dominant in one sex and recessive in the other. Pattern baldness in man is a common example: it is dominant in males but recessive in females.

10. Linkage and crossing over

- **Linkage:** the tendency of genes located on the same chromosome to be inherited together.
- **Crossing over:** the exchange of segments between non-sister chromatids of homologous chromosomes during meiosis, which produces new combinations.

Linked genes on the same chromosome are inherited together and behave almost like a single gene. That is why a dihybrid cross gives a monohybrid-like 3:1 ratio instead of the usual 9:3:3:1 ratio when the gene loci are linked.

11. Epistasis — coat colour in Labrador retriever

An epistatic gene present at one locus suppresses or masks the expression of a gene at a different locus. The masked gene is called the hypostatic gene, and this interaction is called epistasis.

Epistasis is the interaction in which a gene at one locus masks the expression of a gene at another locus. In Labrador retrievers, gene B controls the colour of the pigment: BB or Bb gives black coat and bb gives chocolate (brown) coat. Another gene, E, controls whether the pigment is deposited in the fur at all: E allows deposition, but ee prevents it and the dog is yellow, whatever the B genotype.

Genotype	Coat colour
B_E_	Black
bbE_	Chocolate
B_ee or bbee	Yellow

This gives a 9:3:4 ratio in F2 instead of 9:3:3:1. Here gene E is epistatic over gene B.

Evolution

CHAPTER 10

Evidences of Evolution

LESSON 52

1. Evolution and special creation

Evolution holds that species change gradually over time and new species arise from earlier ones. Special creation, in contrast, holds that species were created in their present form and do not change. Suarez was a supporter of the theory of special creation, according to which all species were created in their present form by a divine power. Buffon, Lamarck and Darwin, in contrast, presented ideas that supported change in species over time.

Special Creation	Evolution
Species were created in their present form and do not change	Species change gradually over time and new species arise from earlier ones

2. Palaeontology — evidence from fossils

Palaeontology, the study of fossils, provides direct historical evidence for evolution. Fossils in older rocks are simpler than those in younger rocks, extinct forms such as dinosaurs are found, and series like the evolution of the horse show step-by-step change from ancestors to modern forms.

- **Older and younger rocks:** fossils in older rocks are simpler than those in younger rocks.
- **Extinct forms:** extinct forms such as dinosaurs are found as fossils.
- **Step-by-step series:** series like the evolution of the horse show gradual change from ancestors to modern forms.

3. Archaeopteryx — a connecting link

Archaeopteryx is a connecting link between reptiles and Aves (birds). It had reptilian features such as teeth, claws and a long bony tail, along with bird features such as feathers and wings. It therefore links reptiles with birds and shows that birds evolved from reptiles.

- **Reptilian features:** teeth, claws and a long bony tail.
- **Bird features:** feathers and wings.

4. Biogeography — evidence from distribution

Biogeography is the study of the distribution of species. Organisms on islands and continents resemble the nearest mainland forms yet differ in a gradual way, such as the finches of the Galapagos Islands, and similar environments on different continents have very different species. This fits descent with modification after continents separated by plate tectonics, not independent creation everywhere. Together, biogeography and palaeontology strongly support evolution.

5. Homologous organs

Homologous organs have the same basic structure and embryonic origin but perform different functions, such as the forelimbs of man, bat, bird and whale. The wings of bat and bird and the flippers of dolphin are all modified vertebrate forelimbs with the same basic bone pattern, so they are homologous.

Such similarity of plan in different groups can only be explained if they inherited the structure from a common ancestor and then modified it for different uses. This strongly supports evolution and divergent change from common ancestry.

6. Analogous organs

Analogous organs are organs that perform the same function but have different structure, origin and ancestry, such as the wings of an insect and the wings of a bird. Both are used for flying, but the insect wing is a chitinous outgrowth without bones, while the bird wing is a modified forelimb with bones. Their similarity is due to convergent evolution under similar conditions, not common ancestry.

The wings of an insect, a bird and a bat perform the same function of flight but have different structure and origin, so they are analogous to each other. The arm of man is not a flight organ and is homologous, not analogous, to the vertebrate wings. The wing of a butterfly is made of chitin and has no vertebrate bones, so it is not homologous with the wings of bat and bird or the flippers of dolphin.

Homologous Organs	Analogous Organs
Same basic structure and embryonic origin, different functions	Same function, different structure, origin and ancestry
Forelimbs of man, bat, bird and whale; wings of bat and bird and flippers of dolphin as modified vertebrate forelimbs	Wings of insect and wings of bird; wings of insect, bird and bat for flight
Show common ancestry and divergent change	Show convergent evolution under similar conditions

7. Convergent and divergent evolution

- **Convergent evolution:** unrelated organisms develop similar features because of similar environments.
- **Divergent evolution:** related organisms develop different features from a common ancestor.

Theories of Evolution — Lamarckism, Darwinism and Neo-Darwinism

LESSON 53

1. Lamarck and Philosophie Zoologique

Jean Baptiste de Lamarck published his theory of evolution in his book Philosophie Zoologique in 1809. In it he proposed the inheritance of acquired characters through use and disuse of organs. Lamarck presented the first systematic theory of organic evolution in this book, known as Lamarckism or the theory of inheritance of acquired characters.

2. Lamarckism — main ideas

- **New needs:** a changing environment creates new needs in organisms.
- **Use of organs:** organs which are used more become stronger and larger; continuously using some organ results in its further strengthening in offspring, in Lamarck's proposal.
- **Disuse of organs:** disused organs weaken and may disappear; continuously disusing some organ results in its weakening in offspring, in Lamarck's proposal.
- **Inheritance of acquired characters:** characters acquired during life are inherited by the offspring.
- **Example:** the long neck of the giraffe, developed by stretching to reach high leaves.

3. Limits of Lamarckism

Lamarck's ideas about use, disuse and strengthening or weakening of organs in the offspring were his proposed explanation, but they were never supported by concrete experimental evidence. Later experiments, such as those of Weismann, showed that characters acquired during life are not inherited. His mechanism was later disproved.

4. Lamarck as an early proponent of evolution

Lamarck was one of the first scientists to propose, in 1809, a complete theory that species are not fixed but change over time to become better adapted to their environment. Although his explanation of inheritance of acquired characters was later rejected, he introduced the ideas of gradual change, adaptation and use and disuse, and prepared the way for Darwin. His contribution remains important because he first established that species change gradually, that adaptation is related to the environment, and that evolution proceeds from simple to complex forms. He is therefore rightly regarded as an early proponent of evolution.

5. Lamarckism and Darwinism

Lamarckism	Darwinism
Explains evolution by use and disuse of organs and inheritance of acquired characters	Explains evolution by variation and natural selection of the fittest

6. Darwinism and genetics

Darwin supported natural selection with evidence from biogeography, comparative anatomy and fossils, but he did not know the mechanism of heredity. Genetics was not developed in his time, so his theory lacked genetic support until Neo-Darwinism combined it with genetics.
Darwinism explains evolution mainly by natural selection acting on variations, but Darwin could not explain the origin and inheritance of variations.

7. Neo-Darwinism

Neo-Darwinism is the modern form of the theory that adds genetics. Variations arise by mutation and recombination, are stored in the gene pool of a population, and natural selection, genetic drift and isolation change allele frequencies to produce new species.

Darwinism	Neo-Darwinism
Explains evolution mainly by natural selection acting on variations; origin and inheritance of variations not explained	Adds genetics — variations arise by mutation and recombination and are stored in the gene pool
No genetic support in Darwin's time	Natural selection, genetic drift and isolation change allele frequencies to produce new species

Darwin's Theory of Natural Selection

LESSON 54

1. Malthus and the struggle for existence

Thomas Malthus proposed that human population increases faster than its food supply, so many individuals must die in the struggle for existence. Darwin applied this idea to all species and developed the idea of survival of the fittest through natural selection. The theory on human population which inspired Darwin was proposed by Malthus.

2. Observations behind natural selection

- **Overproduction:** organisms produce far more offspring than can survive.
- **Limited resources:** resources are limited, so there is a struggle for existence.
- **Variation:** individuals of a population show heritable variations.
- **Unequal fitness:** organisms are not all equally fit for their environment.

3. Theory of natural selection

Individuals with variations favourable for that environment survive and reproduce more successfully — survival of the fittest — and pass their useful characters to their offspring, while the less fit are gradually eliminated. Continued over many generations, this natural selection accumulates favourable variations and produces new, better adapted species.

Chain: overproduction → struggle for existence → heritable variation → survival of the fittest → useful characters passed to offspring → accumulation over many generations → new, better adapted species.

4. Descent with modification

Descent with modification is Darwin's phrase for evolution: present-day species have descended, with gradual changes, from pre-existing ancestral species. Over long periods these small inherited modifications accumulate, producing organisms that are different from their ancestors and, in time, entirely new species. The process of natural selection producing new species is the process Darwin called descent with modification.

5. Alfred Wallace and natural selection

Alfred Russel Wallace, working independently in the Malay Archipelago, reached the same idea of natural selection as Darwin and sent his essay to Darwin in 1858. This contribution pushed Darwin to finally publish his work, and their papers were presented together, after which Darwin published On the Origin of Species in 1859. Wallace also contributed greatly through his studies of the geographical distribution of animals.

Event	Detail
Wallace's work	Reached the same idea of natural selection independently in the Malay Archipelago
1858	Wallace sent his essay to Darwin; their papers were presented together
1859	Darwin published On the Origin of Species

Population Genetics — Hardy-Weinberg Equilibrium and Genetic Drift

LESSON 55

1. Hardy-Weinberg equilibrium

The Hardy-Weinberg equilibrium stays undisturbed when mating is random, there is no selection, no mutation, no migration and the population is large. Selection, non-random mating and mutation change allele frequencies, so they disrupt the equilibrium, while random mating does not. Random mating therefore cannot be a disruptive force for the Hardy-Weinberg theorem.

- **Conditions for equilibrium:** random mating, no selection, no mutation, no migration and a large population.
- **Disruptive forces:** selection, non-random mating and mutation change allele frequencies and disrupt the equilibrium.

2. Factors affecting allele frequency

The main factors that change allele frequency in a population are mutation, natural selection, genetic drift, gene flow and non-random mating. Gene flow is the migration of individuals in or out of the population. Any of these factors disturbs the Hardy-Weinberg equilibrium and can lead to evolution of the population over generations.

- **Mutation**
- **Natural selection**
- **Genetic drift**
- **Gene flow:** migration of individuals in or out of the population.
- **Non-random mating**

3. Genetic drift

Genetic drift is the random change in allele frequencies in a population from one generation to the next, occurring purely by chance. It is strongest in small populations, where a few individuals breeding or dying by chance can greatly change or even eliminate an allele, as seen in the bottleneck and founder effects.

4. Bottleneck effect and founder effect

Bottleneck Effect	Founder Effect
A disaster greatly reduces the size of a population and its genetic variation	A small group separates from a large population and starts a new population with a different gene pool

Both effects are examples of genetic drift, in which allele frequencies change by chance, with the strongest effects in small populations.

Origin of Life, Origin of Eukaryotes and Speciation

LESSON 56

1. Origin of life in water

The first life forms originated in water, especially in shallow seas and hot springs called hydrothermal vents. Water provided the medium in which simple chemicals combined to form complex organic compounds and the first cells.

2. Origin of single-celled eukaryotes

Single-celled eukaryotes arose from prokaryotic ancestors more than about 1.5 billion years ago in water. These changes gave the first single-celled eukaryotes a nucleus, organelles and greater efficiency, from which all later eukaryotic life evolved.

3. Invagination theory

According to the invagination theory, the cell membrane of a prokaryote folded inwards and enclosed the genetic material, forming a true nucleus and internal membranes such as the endoplasmic reticulum. The invagination theory says eukaryotic organelles formed when the prokaryotic cell membrane folded inwards to enclose the genetic material.

4. Endosymbiotic theory

According to the endosymbiotic theory, larger prokaryotic cells engulfed smaller aerobic bacteria, which were not digested but lived inside them in symbiosis and became mitochondria, and photosynthetic cyanobacteria similarly became chloroplasts. The endosymbiotic theory says mitochondria and chloroplasts arose from free-living bacteria that were taken inside another cell and lived in symbiosis.

Invagination Theory	Endosymbiotic Theory
Eukaryotic organelles formed when the prokaryotic cell membrane folded inwards to enclose the genetic material; this formed a true nucleus and internal membranes such as the endoplasmic reticulum	Mitochondria and chloroplasts arose from free-living bacteria taken inside another cell and living in symbiosis; aerobic bacteria became mitochondria and photosynthetic cyanobacteria became chloroplasts

5. Speciation

Speciation is the biological process by which one or more new species are formed from an existing species when populations become reproductively isolated and accumulate genetic differences.

6. Allopatric speciation

In allopatric speciation, a geographical barrier such as a river, mountain or sea separates a population, and the separated parts evolve independently into new species.

7. Sympatric speciation

In sympatric speciation a new species arises within the same geographical area, without any physical barrier, often through polyploidy or a change in habits or mating time. A new species arises in the same area, for example by polyploidy in plants or a change in food, habitat or breeding season.

8. Parapatric speciation

In parapatric speciation a new species arises in a population that lives in neighbouring, partly overlapping areas, where gene flow is limited and the two ends of the range gradually become different species. Neighbouring populations in adjoining areas with limited gene flow gradually diverge into separate species.

Type	How a New Species Arises
Allopatric	A geographical barrier such as a river, mountain or sea separates a population; the separated parts evolve independently
Sympatric	Within the same geographical area, without a physical barrier, often through polyploidy or a change in habits or mating time
Parapatric	In neighbouring, partly overlapping areas, where gene flow is limited and the two ends of the range gradually become different species

Man and his Environment

CHAPTER 11

Ecosystem Producers and Ecological Succession

LESSON 57

1. Major producers in a terrestrial ecosystem

The major producers in a terrestrial ecosystem are green plants, including grasses, herbs, shrubs and trees, which make food by photosynthesis. All other organisms of the ecosystem depend directly or indirectly on them for food and energy.

- **In a forest:** the trees are the chief producers.
- **In grassland:** the grasses play the role of chief producers.

2. Ecological succession

Succession is the gradual and directional process by which the community of an area changes over time until a stable climax community is formed. Simple pioneer species are replaced step by step by more complex and larger species through seral stages.

Succession increases the diversity, biomass and stability of the community and finally produces a climax community that is in balance with the climate of the area.

3. Pioneers in xerarch succession

Xerarch succession starts on dry bare rock. Crustose lichens are the first, pioneer organisms to grow there. They slowly break the rock surface and help in soil formation, preparing the way for mosses and higher plants.

4. Secondary succession

Secondary succession occurs where a community has been disturbed but soil and some organisms remain, for example a burnt forest or an abandoned cultivated field. Because soil is already present, secondary succession is much faster than primary succession.

- **Stage 1:** annual weeds and grasses grow first.
- **Stage 2:** perennial herbs and shrubs appear.
- **Stage 3:** fast-growing softwood trees such as pine appear.
- **Final stage:** hardwood trees form the climax forest.

Chain: disturbed area with soil remaining → annual weeds and grasses → perennial herbs and shrubs → softwood trees such as pine → hardwood climax forest.

Biogeochemical Cycles — Meaning and Types

LESSON 58

1. What a biogeochemical cycle is

A biogeochemical cycle is the repeated circulation of an element, such as carbon or nitrogen, between living organisms and the non-living environment, so that the element is used again and again.

2. Why it is called biogeochemical

It is called biogeochemical because the elements involved move in a cycle through living organisms (bio), the Earth and soil (geo) and chemical changes in air and water (chemical). For example, nitrogen passes from air to soil, into plants and animals, and back to air through the activities of living organisms and chemical processes.

- **Bio:** living organisms.
- **Geo:** the Earth and soil.
- **Chemical:** chemical changes in air and water.

3. Gaseous and sedimentary cycles

Carbon, nitrogen and oxygen cycles are gaseous cycles because their main reservoir is the atmosphere or oceans. The phosphorus cycle is not a gaseous type of cycle: in the phosphorus cycle the main reservoir is rocks and soil, not the atmosphere, so it is a sedimentary cycle.

Cycle	Type	Main reservoir
Carbon cycle	Gaseous	Atmosphere or oceans
Nitrogen cycle	Gaseous	Atmosphere or oceans
Oxygen cycle	Gaseous	Atmosphere or oceans
Phosphorus cycle	Sedimentary	Rocks and soil

4. Source of carbon to plants

The source of carbon to plants in the carbon cycle is atmospheric carbon dioxide. Green plants take carbon dioxide directly from the atmosphere for photosynthesis and convert it into food. From plants this carbon passes to animals and other organisms through food chains.

The Nitrogen Cycle

LESSON 59

1. The nitrogen cycle in detail

Atmospheric nitrogen is about 78 percent of air. It is first fixed into compounds by nitrogen-fixing bacteria in soil and in the root nodules of legumes, by lightning and by industry, forming ammonia and nitrates.

- **Nitrogen fixation:** atmospheric nitrogen is fixed into compounds by nitrogen-fixing bacteria in soil and in the root nodules of legumes, by lightning and by industry, forming ammonia and nitrates.
- **Use by plants and animals:** plants absorb nitrates and make proteins, and animals obtain nitrogen by eating plants.
- **Ammonification:** the dead bodies and wastes of organisms are broken down by decomposers into ammonia.
- **Nitrification:** nitrifying bacteria convert ammonia into nitrites and nitrates, which plants absorb again.
- **Denitrification:** denitrifying bacteria convert some nitrates back into free nitrogen gas, completing the cycle.

Chain: atmospheric nitrogen → fixation → ammonia and nitrates → plant proteins → animals → dead bodies and wastes → ammonia by ammonification → nitrites and nitrates by nitrification → plants again, with some nitrates returned to free nitrogen by denitrification.

2. Nitrification and denitrification compared

Nitrification	Denitrification
Conversion of ammonia into nitrites and then nitrates by nitrifying bacteria such as Nitrosomonas and Nitrobacter	Denitrifying bacteria convert nitrates back into free nitrogen gas, which escapes into the atmosphere
Makes nitrogen available to plants in the soil	Reduces soil fertility

Denitrification is the conversion of nitrates to nitrogen. It is the reverse of nitrification, in which ammonia is converted into nitrites and nitrates.

3. Nitrogen balance and proteins

Nitrogen enters the body mainly through proteins in food and is lost mainly in urea in urine. Nitrogen balance compares this intake and loss of nitrogen, so it depends directly on the amount of protein eaten. The dietary requirement involved in nitrogen balance is therefore proteins.

Energy Flow Through Food Chains and Food Webs

LESSON 60

1. The Sun — source of energy for ecosystems

Solar energy is the ultimate and only original source of energy for almost all ecosystems on Earth. Producers capture it by photosynthesis, and from them energy flows to all consumers through food chains.

2. How energy flows between trophic levels

Energy flows in one direction only, from producers to herbivores, then to carnivores and finally to decomposers, through food chains and food webs.

Chain: producers → herbivores → carnivores → decomposers.

- **Passed on:** at each transfer only about 10 percent of the energy is passed on.
- **Lost or used:** the rest is lost as heat, used in life activities or left in undigested material.
- **Result:** higher trophic levels receive much less energy.

3. Heat loss at each trophic level

Heat loss is a factor that affects the flow of energy at the trophic level. At every trophic level a large part of the energy is lost as heat during respiration and other life activities. Because of this heat loss, only about 10 percent of the energy of one trophic level passes to the next level.

At one trophic level	Share of energy
Passed to the next trophic level	About 10 percent
Lost as heat, used in life activities, or left in undigested material	The rest

4. Processes that absorb heat energy

Melting of ice and evaporation of water absorb heat energy from the surroundings, so the energy change for these processes is positive (endothermic). Such processes take in energy instead of releasing it.

Global Warming, Ozone Depletion, Population and Conservation

LESSON 61

1. The greenhouse effect and global warming

The greenhouse effect is the trapping of heat in the atmosphere by greenhouse gases such as carbon dioxide, and its increase is seen as global warming. As the Earth becomes warmer, ice melts, sea level rises and weather patterns change.

2. Rise of carbon dioxide in the atmosphere

The main cause of the increase in the amount of carbon dioxide in the atmosphere is the burning of a large number of fossil fuels. Burning of coal, oil and gas in vehicles, factories and power houses releases huge amounts of carbon dioxide into the air. This is the main cause of the steady rise of carbon dioxide in the atmosphere and of the greenhouse effect.

3. Effects of carbon dioxide and global warming on human health

- **Heat illness:** higher temperatures cause heat strokes and dehydration, especially in old people and outdoor workers.
- **Infectious diseases:** warmer and wetter conditions help mosquitoes and other vectors spread, increasing malaria, dengue and other infectious diseases.
- **Food and water:** global warming disturbs rainfall and agriculture, causing food shortages and malnutrition.
- **Displacement:** melting ice and rising sea level displace coastal populations.
- **Breathing problems:** increased air pollution and allergens along with warming worsen asthma and other breathing problems.

4. Causes of ozone layer depletion

The main cause of ozone layer depletion is the release of chlorofluorocarbons (CFCs) used in refrigerators, air conditioners, sprays and foams. Other ozone-destroying substances include halons, carbon tetrachloride and nitrogen oxides from jet planes and fertilisers. In the upper atmosphere these release chlorine, which breaks down ozone molecules and thins the protective ozone layer.

5. Factors that determine the growth rate of the human population

The growth rate of the human population is determined mainly by birth rate and death rate, and also by immigration and emigration. Improvements in agriculture, industry and control of diseases are the reasons why death rate has fallen much faster than birth rate, causing rapid population growth.

Factor	Effect and reason
Birth rate	A high birth rate increases population because more children are added
Death rate	Better medical facilities, sanitation and food supply have reduced the death rate, so people live longer and population grows faster
Immigration	Increases the population of the area people move into
Emigration	Decreases the population of the area people leave

6. Why biodiversity should be conserved

Biodiversity provides food, medicines, fibres and many other useful products and maintains ecosystem services such as pollination, soil formation, water purification and balance of gases in the air. Loss of species can collapse food webs and make ecosystems unstable, so conserving biodiversity is essential for human survival and well-being.

7. Renewable and non-renewable resources

Renewable resources	Non-renewable resources
Replaced naturally in a short time and can be used again and again if used carefully	Present in fixed amounts and are not replaced after use within a human lifetime
Examples: sunlight, water, air, forests and animals	Examples: coal, oil, natural gas and minerals
Can be used again and again if used carefully	Must be used wisely and conserved

Biotechnology

CHAPTER 12

Genetic Engineering and Recombinant DNA Technology

LESSON 62

1. Gene cloning

Cloning of genes creates an identical copy of a DNA sequence. Cloned genes provide many identical copies of a gene.

- **Transgenic organisms:** cloned genes can be inserted into organisms to create transgenic organisms.
- **Diagnosis:** cloned genes can be used as probes to diagnose genetic disorders.
- **Study of gene function:** cloning allows scientists to study the function of a gene in detail.

2. Marker genes in the construction of recombinant DNA

AmpR and LacZ are used as selectable and screenable marker genes in the plasmid vector during construction of recombinant DNA.

- **AmpR:** gives resistance to the antibiotic ampicillin, so only bacteria that have taken up the plasmid can grow on ampicillin medium.
- **LacZ:** produces an enzyme that gives blue colonies. When foreign DNA is inserted into LacZ, the gene is inactivated and the colonies stay white.
- **Blue-white screening:** recombinant bacteria can be recognised by blue-white screening — blue colonies have an active LacZ gene, while white colonies carry the inserted foreign DNA.

Marker gene	Type	How it works
AmpR	Selectable marker	Gives ampicillin resistance; only bacteria that have taken up the plasmid grow on ampicillin medium
LacZ	Screenable marker	Active gene gives blue colonies; insertion of foreign DNA inactivates it and colonies stay white

3. Enzymes used in recombinant DNA technology

- **Restriction endonucleases:** cut DNA at specific sites.
- **DNA ligase:** joins DNA fragments.
- **DNA polymerase:** synthesises new DNA strands.
- **Alkaline phosphatase:** used to prevent a cut vector from rejoining to itself.
- **Taq polymerase:** used in PCR to amplify DNA.

Enzyme	Job in recombinant DNA technology
Restriction endonucleases	Cut DNA at specific sites
DNA ligase	Joins DNA fragments
DNA polymerase	Synthesises new DNA strands
Alkaline phosphatase	Prevents a cut vector from rejoining to itself
Taq polymerase	Amplifies DNA in PCR

Tools of Biotechnology — Restriction Enzymes, PCR, Gel Electrophoresis and Blotting

LESSON 63

1. Restriction enzymes as molecular scissors

Restriction enzymes cut DNA at specific palindromic sequences, just as scissors cut at a chosen point. Each enzyme recognises its own specific sequence and cuts both strands there, often producing sticky ends. Because they cut DNA molecules precisely at desired sites for joining with other DNA, they are called molecular scissors.

2. Palindromic sequences in DNA

A palindrome in DNA is a sequence of bases that reads the same on both strands when read in the 5 to 3 direction, like the word MADAM reads the same from both sides.

- **Example:** the sequence GAATTC on one strand pairs with CTTAAG on the other and reads GAATTC in both directions.
- **Recognition sites:** restriction enzymes recognise and cut such palindromic sequences.

3. Polymerase chain reaction — PCR

PCR is a technique used to amplify DNA. It makes millions of copies of a small DNA sample, providing enough DNA for sequencing.

- **Sequencing:** the amplified DNA provides enough material for sequencing.
- **Identifying genetic disorders:** the amplified DNA is used to identify genetic disorders.
- **Gene expression:** in its reverse-transcriptase form, PCR is used to study gene expression.

4. Gel electrophoresis

Gel electrophoresis is a technique used to separate DNA fragments based on size. DNA fragments move through the gel at different speeds according to their size, forming bands that can be compared between samples. Differences in band patterns reveal genetic variations, which is used in DNA fingerprinting and diagnosis.

5. Southern blotting

Southern blotting is a technique used to transfer DNA fragments from a gel to a membrane. DNA fragments fixed on a membrane are detected with a labelled probe that binds to a specific sequence. This detects mutations and abnormal gene fragments, so it is used to identify genetic disorders.

Technique	What it does	Use given here
PCR	Makes millions of copies of a small DNA sample	Provides DNA for sequencing; identifies genetic disorders; reverse-transcriptase form studies gene expression
Gel electrophoresis	Separates DNA fragments according to size into bands	Differences in band patterns reveal genetic variations; used in DNA fingerprinting and diagnosis
Southern blotting	Transfers DNA fragments from a gel to a membrane, where a labelled probe binds to a specific sequence	Detects mutations and abnormal gene fragments; identifies genetic disorders

DNA Sequencing, DNA Analysis and Genomic Maps

LESSON 64

1. DNA sequencing

DNA sequencing is the technique of determining the exact order of nucleotides (A, T, G and C) in a DNA molecule. By reading the exact order of nucleotides, sequencing reveals changes or mutations in DNA when a sequence is compared with a normal reference sequence. These identified variations help in diagnosing genetic diseases and in studying differences between individuals and species.

2. Sanger sequencing

Sanger sequencing determines the exact order of nucleotides in a DNA fragment by using chain-terminating nucleotides. It was the standard method used to sequence genes and even whole genomes, including in the Human Genome Project. DNA sequencing is most commonly done by the Sanger method, using chain-terminating nucleotides and gel electrophoresis.

3. Sequencing and the study of genetic mutations and diseases

To study a disease, the gene of a patient is sequenced and compared with the normal sequence; any change, such as substitution, deletion or insertion of bases, identifies the mutation causing the disorder, as in cystic fibrosis, sickle cell anaemia and many cancers.

- **Carrier detection:** sequencing has allowed carrier detection.
- **Prenatal diagnosis:** sequencing has allowed prenatal diagnosis.
- **Disease genes:** sequencing allowed discovery of disease genes through the Human Genome Project.
- **Treatment:** sequencing allows the choice of suitable treatment for individual patients.

4. DNA analysis

DNA analysis examines a sample of DNA to detect variations, mutations or specific markers present in it. Finding such variations makes it possible to identify genetic disorders, carriers of a disease and relationships between individuals.

5. Genomic maps

A genomic map shows the positions of genes and other markers on all the chromosomes of an organism. By comparing maps of healthy and diseased persons, scientists can locate and identify the genes responsible for specific diseases.

6. Creating a genomic map

- **Genetic linkage map:** created first, in which the distance between markers is estimated from how often they are separated by crossing over in families.
- **Physical map:** created next, in which overlapping DNA fragments are cloned, cut with restriction enzymes and sequenced to fix their exact positions in base pairs.

Genomic maps, completed for man in the Human Genome Project, have been used to locate disease genes such as those for cystic fibrosis and Huntington disease, to identify carriers, to compare genomes of different species and to develop new diagnostic tests and gene-based medicines.

Type of genomic map	How it is made
Genetic linkage map	Distance between markers estimated from how often they are separated by crossing over in families
Physical map	Overlapping DNA fragments are cloned, cut with restriction enzymes and sequenced to fix their exact positions in base pairs

Applications in Medicine — Medicines, Gene Therapy and Diagnosis

LESSON 65

1. Studying gene function as the basis of healthcare

Studying how genes work and how faulty genes cause disease is the basis of the healthcare uses of biotechnology. This knowledge makes gene therapy, production of drugs such as insulin, and accurate molecular diagnosis of diseases possible. Biotechnology has many applications in healthcare, including gene therapy, drug production, and diagnosis of diseases.

2. Uses of biotechnology in healthcare

- **Medicines from transgenic bacteria:** producing medicines such as insulin, growth hormone and clotting factors in transgenic bacteria.
- **Gene therapy:** one of the healthcare uses of biotechnology.
- **Vaccines:** one of the healthcare uses of biotechnology.
- **Monoclonal antibodies:** one of the healthcare uses of biotechnology.
- **Molecular diagnosis:** molecular diagnosis of diseases before symptoms appear.

3. Benefits and challenges in healthcare

Benefits	Challenges
Safer and cheaper medicines	High cost
Accurate early diagnosis	Ethical questions about altering genes
Treatment of previously incurable genetic diseases	Possible side effects
	Risk of misuse of genetic information

4. Monoclonal antibodies in diagnosis

Monoclonal antibodies are identical antibodies that bind to one specific antigen, so they detect that substance very accurately. They are widely used in diagnostic tests, for example pregnancy tests and tests for infections and cancers.

- **Specific binding:** each monoclonal antibody binds to one specific antigen.
- **Diagnostic tests:** pregnancy tests and tests for infections and cancers.

DNA Analysis in Forensic Science and Paternity Testing — RFLPs and Fingerprinting

LESSON 66

1. Identifying a person from DNA

DNA analysis identifies a person from a very small sample such as blood, hair, semen or skin cells, because the pattern of DNA markers of every person (except identical twins) is unique.

2. DNA analysis in forensic science

In forensic science, DNA fingerprinting of evidence from a crime scene is matched with the DNA of suspects to identify the criminal or to prove an innocent person free.

3. DNA analysis in paternity testing

In paternity testing the markers of a child are compared with those of the mother and the alleged father, since half of the child's markers must come from the father.

4. Methods used in DNA analysis

- **Extraction:** extraction of DNA.
- **Amplification:** amplification of DNA by PCR.
- **Cutting:** cutting with restriction enzymes to obtain RFLPs.
- **Separation:** separation by gel electrophoresis.
- **Comparison:** comparison of STR (short tandem repeat) patterns.

Chain: extraction of DNA → amplification by PCR → cutting with restriction enzymes to obtain RFLPs → separation by gel electrophoresis → comparison of STR patterns.

5. RFLPs — restriction fragment length polymorphisms

RFLPs are differences in the lengths of DNA fragments produced when DNA of different individuals is cut with the same restriction enzyme. These differences arise because mutations change the cutting sites of the enzyme from person to person.

- **Genetic markers:** RFLP patterns are used as genetic markers.
- **Uses:** RFLP patterns are used in DNA fingerprinting, disease diagnosis and mapping of genes.

Application	How DNA analysis is used
Forensic science	DNA fingerprinting of crime-scene evidence is matched with the DNA of suspects to identify the criminal or to prove an innocent person free
Paternity testing	Markers of a child are compared with those of the mother and the alleged father; half of the child's markers must come from the father

Biology and Human Welfare

CHAPTER 13

Role of Microorganisms in Human Welfare

LESSON 67

1. Microbes in human welfare

Microbes are used in food processing and in sewage treatment. In both fields microbes convert waste or raw material into useful, safe products.

2. Microbes in food processing

Microbes are used to make many everyday foods, improving their taste, texture and keeping quality.

- **Lactobacillus:** converts milk into curd and yoghurt.
- **Saccharomyces cerevisiae:** the yeast used to raise bread dough and to produce alcohol.
- **Other microbes:** used in cheese, vinegar and fermented foods.

3. Yeast and the puffy appearance of dough

The yeast Saccharomyces cerevisiae ferments sugars in the dough and releases carbon dioxide gas. The trapped gas bubbles make the dough rise and give it a puffy, spongy appearance, which is used in making bread.

- **Fermentation:** yeast ferments sugars in the dough.
- **Carbon dioxide:** gas released during fermentation is trapped as bubbles in the dough.
- **Result:** the dough rises and becomes puffy and spongy.

4. Microbes in sewage treatment

In sewage treatment, microbes do the main cleaning work. Aerobic bacteria and other microorganisms in secondary treatment decompose the organic pollutants of sewage into harmless substances, and anaerobic bacteria digest the sludge, producing methane gas that can be used as fuel.

5. Steps of sewage treatment

Sewage treatment is done in three main steps: primary, secondary and tertiary treatment.

- **Primary treatment:** the physical first stage, in which sewage is passed through screens and sedimentation tanks and floating and suspended solids are removed by screening, filtration and sedimentation. The settled solid material forms sludge, the clear liquid is called effluent, and the liquid passes on for biological treatment.
- **Secondary treatment:** a biological process in which air is bubbled through the effluent and aerobic microbes decompose the dissolved organic matter, greatly reducing its harmful content. The microbes then settle as activated sludge, part of which is used again and part sent for sludge digestion.
- **Tertiary treatment:** the water is further purified by chemical methods and disinfection, for example with chlorine or ultraviolet light, to remove remaining nutrients, pathogens and chemicals before the water is released or reused.

Stage	Type of process	Main work
Primary treatment	Physical	Floating and suspended solids removed by screening, filtration and sedimentation; settled solids form sludge, clear liquid is effluent
Secondary treatment	Biological	Air bubbled through effluent; aerobic microbes decompose dissolved organic matter; microbes settle as activated sludge
Tertiary treatment	Chemical and disinfection	Remaining nutrients, pathogens and chemicals removed, for example with chlorine or ultraviolet light, before release or reuse

6. Sludge digestion

Sludge digestion is a biological process in which organic solids are decomposed into stable substances. During sludge digestion, anaerobic microbes break down the organic matter of sludge into stable, harmless substances, along with gases such as methane. Part of the activated sludge is sent for sludge digestion by anaerobic bacteria, which also produce biogas. The digested sludge can then be dried and safely used as manure.

- **Anaerobic microbes:** break down the organic matter of sludge.

- **Stable substances:** the harmless end products of digestion.
- **Methane / biogas:** gas produced during digestion, which can be used as fuel.
- **Digested sludge:** can be dried and safely used as manure.

Food Production — Livestock, Milk Breeds and Vegetables

LESSON 68

1. Food production from livestock

Livestock and dairy animals are a major part of food production. They provide highly nutritious food and many other products for human welfare.

2. Role of livestock

Livestock animals such as cow, buffalo, goat, sheep and poultry provide highly nutritious food in the form of milk, meat and eggs, which supply proteins, fats and vitamins.

- **Food:** milk, meat and eggs, supplying proteins, fats and vitamins.
- **Other products:** wool, leather and other products.
- **Draught power:** power for farming and transport.
- **Manure:** manure for fields.
- **Income and employment:** livestock is a major source of income and employment for rural families and an important part of the national economy.

3. Top milk producing animal of Pakistan

The Nili-Ravi buffalo, a famous breed of Punjab, is the top milk producing animal of Pakistan. A good animal gives a high daily yield of milk with high fat content, so it is most valued by dairy farmers.

- **Nili-Ravi:** buffalo breed of Punjab; top milk producing animal of Pakistan, with high daily yield and high fat content.

4. Outstanding milk producing breeds of cow

Outstanding milk producing breeds of cow include local breeds and exotic high-yielding breeds, and crosses of the two.

- **Sahiwal:** a local breed, well adapted to the hot climate of Pakistan and giving good yields.
- **Red Sindhi:** a local breed, well adapted to the hot climate of Pakistan and giving good yields.
- **Holstein Friesian:** an exotic breed and the highest milk producer.
- **Jersey:** an exotic breed kept for rich, high-fat milk.
- **Brown Swiss:** an exotic breed kept for rich, high-fat milk.
- **Crosses:** crosses of local breeds with exotic breeds combine high yield with adaptation to local conditions.

Group	Breeds	Main feature
Local cow breeds	Sahiwal, Red Sindhi	Well adapted to the hot climate of Pakistan, good yields
Exotic cow breeds	Holstein Friesian, Jersey, Brown Swiss	High-yielding; Holstein Friesian is the highest milk producer; Jersey and Brown Swiss give rich, high-fat milk
Buffalo	Nili-Ravi	Top milk producing animal of Pakistan; high daily yield with high fat content

5. Winter and summer vegetables

French beans and cauliflower grow best in cool weather, so in Pakistan they are sown and harvested as winter vegetables. High summer heat spoils their growth and, in cauliflower, prevents proper curd formation.

- **Winter vegetables:** French beans and cauliflower, growing best in cool weather.
- **Effect of heat:** high summer heat spoils growth and, in cauliflower, prevents proper curd formation.

Disease Control — Vaccination, Vaccines and Integrated Management

LESSON 69

1. Vaccination and protection against infectious disease

Vaccination introduces a weakened or killed pathogen, or its toxin, into the body, so the immune system makes antibodies and memory cells against it without causing the disease. When the real pathogen later attacks, the body responds quickly and the person does not become ill.

2. What vaccination is

Vaccination is the process of introducing a vaccine, made from weakened or killed microbes or their inactivated toxins, into the body to protect it against a specific disease. The method was first developed by Edward Jenner in 1796 using cowpox against smallpox. The vaccine acts as an antigen: the body produces antibodies and memory cells, so on later natural infection the immune response is rapid and strong and the disease is prevented.

- **Vaccine:** made from weakened or killed microbes or their inactivated toxins.
- **Antigen action:** the vaccine acts as an antigen and stimulates antibody and memory cell formation.
- **Later infection:** the immune response is rapid and strong, and the disease is prevented.
- **Edward Jenner:** first developed vaccination in 1796, using cowpox against smallpox.

3. Importance of vaccination

Vaccination of a whole population has controlled or eradicated dangerous infectious diseases such as smallpox and polio. Vaccination has eradicated smallpox, has nearly eradicated polio, and controls diseases such as measles, diphtheria, tetanus, hepatitis and COVID-19, saving millions of lives, especially of children, and reducing the cost of treatment.

- **Eradicated:** smallpox.
- **Nearly eradicated:** polio.
- **Controlled:** measles, diphtheria, tetanus, hepatitis and COVID-19.
- **Benefit:** millions of lives saved, especially of children, and reduced cost of treatment.

4. DT vaccine and diphtheria

DT vaccine contains diphtheria (and tetanus) toxoid, that is the inactivated toxin of the bacteria. It stimulates the body to make antibodies that protect against diphtheria.

- **DT vaccine:** contains diphtheria and tetanus toxoid.
- **Toxoid:** the inactivated toxin of the bacteria.
- **Protection:** stimulates the body to make antibodies against diphtheria.

5. Tetanus and Clostridium tetani

Clostridium tetani, which causes tetanus, is a bacterium. It is a rod-shaped bacterium found in soil and dust that enters the body through wounds. Its toxin affects the nervous system and causes the painful muscle stiffness of tetanus.

- **Clostridium tetani:** a rod-shaped bacterium, not a virus, parasite or fungus.
- **Habitat and entry:** found in soil and dust; enters the body through wounds.
- **Toxin effect:** affects the nervous system and causes painful muscle stiffness.

6. MMR vaccine

MMR is a combined vaccine whose name is made from the first letters of the three viral diseases it prevents: measles, mumps and rubella. It is given to children to provide immunity against all three diseases together.

Vaccine	Protects against	Key fact
DT	Diphtheria (and tetanus)	Contains diphtheria and tetanus toxoid, the inactivated bacterial toxin
MMR	Measles, mumps and rubella	Combined vaccine for three viral diseases, given to children

7. Integrated disease management

Integrated disease management means the planned control of the serious and common diseases of a population by using all suitable measures together, instead of depending on one method alone. Used together, these measures reduce the spread and severity of diseases effectively and economically.

- **Vaccination:** one of the combined measures.
- **Healthy habits and sanitation:** part of the combined approach.
- **Clean water and food:** part of the combined approach.
- **Early diagnosis and treatment:** part of the combined approach.
- **Biological control:** control using other living organisms.

Breeding Methods and Genetic Engineering

LESSON 70

1. Improving organisms for human welfare

Selection, hybridisation, back cross and genetic engineering are methods used to improve plants and animals and to enhance useful characteristics.

2. Back cross

When an F1 hybrid is crossed with either of its parents, the cross is called a back cross. Breeders use it to transfer a desired character of the hybrid into the parent variety while keeping most of the parent's good qualities.

- **Back cross:** crossing of an F1 hybrid with either of its parents.
- **Purpose:** to transfer a desired character of the hybrid into the parent variety while keeping most of the parent's good qualities.

3. Selection, back cross and hybridisation compared

Selection is the choosing of plants or animals with desired characters for breeding. Selection improves a breed by repeatedly breeding from the best existing individuals of the same variety, while hybridisation produces a new hybrid by crossing two different varieties or breeds to combine their desired characters.

Method	Meaning
Selection	Choosing plants or animals with desired characters for breeding; improves a breed by repeatedly breeding from the best existing individuals of the same variety
Back cross	Crossing of an F1 hybrid with one of its parents
Hybridisation	Produces a new hybrid by crossing two different varieties or breeds to combine their desired characters

4. Vegetables of summer and vegetables of winter

Group	Examples	Growing conditions and sowing
Summer vegetables	Okra (bhindi), bitter gourd, bottle gourd and brinjal	Grow in hot weather and are sown in spring or summer
Winter vegetables	Cauliflower, cabbage, carrot, radish and peas	Grow in cool weather and are sown in autumn or winter

French beans and cauliflower also belong with the winter vegetables, as they grow best in cool weather.

5. Genetic engineering

Genetic engineering is the science that deals with the modification of DNA and enhances the characteristic of an organism. Genetic engineering directly changes the DNA of an organism by removing, altering or adding genes. In this way a desired characteristic is enhanced or a new useful characteristic is introduced into the organism.

- **Genetic engineering:** direct change to the DNA of an organism by removing, altering or adding genes.
- **Result:** a desired characteristic is enhanced, or a new useful characteristic is introduced.